

# Faroese Whole Genomes Provide Insight into Ancestry and Recent Selection

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## eLife Assessment

This paper presents an analysis of demography and selection from whole-genome sequencing of 40 Faroese, with data that are **useful** beyond the study region. Much of the analysis is **solid**, but a more fine-scale analysis of demographic history could have led to more interesting findings. In addition, there are concerns about the selection analyses, given the special nature of the studied population and sampling scheme. Finally, lack of data availability limits the broader value of the paper.

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## Abstract

The Faroe Islands are home to descendants of a North Atlantic founder population with a unique history shaped by both migration and periods of relative isolation. Here, we investigate the genetic diversity, population structure, and demographic history of the islands by analyzing whole genome sequencing data from 40 participants in the Faroe Genome Project. This represents the first whole genome sequencing panel of this size from the Faroe Islands. We observed numerous putatively functional private alleles, including stop gain variants and high impact missense variants in the cohort. Faroese individuals had a higher proportion of their genomes contained in long runs of homozygosity than other European groups, including Finnish, suggesting a more recent or stronger bottleneck in the Faroese population. Signals of positive selection were identified at loci containing genes that play roles in vitamin D and dietary fat absorption and DNA repair, while increased diversity on lactase persistence haplotypes was observed. Fine-scale analysis of haplotype structure in modern and ancient European genomes revealed genetic affinities with ancient Iron Age individuals from the North and West of Europe, providing evidence for potential contributions to the Faroese gene pool from Celtic and Viking populations as well as

information about the temporal order in which these events happened. This study highlights the impact of evolutionary processes, such as ancient admixture, founder events, and positive selection, on the present-day genetic architecture of North Atlantic founder populations like the Faroe Islands.

## Introduction

The Faroe Islands, nestled in the North Atlantic Ocean between Iceland and Norway, are home to the descendants of a North Atlantic founder population with a rich cultural heritage and a unique history shaped by both migration and periods of relative isolation. The exact settlement history of the islands is unclear, though historical records and analysis of Y-chromosome microsatellite markers point to a few founders most likely having arrived primarily from Scandinavia and the British Isles starting around the 9th century C.E.<sup>1,2,3</sup> However, archeological evidence supports a possible earlier settlement of the island around the 4th-6th centuries C.E.<sup>4</sup> Studies of mtDNA reveal an excess of maternal ancestry from the British Isles, while Y chromosome studies reveal an excess of paternal ancestry from Scandinavia, suggesting sex- biased admixture between these ancestral groups during the founding of the population.<sup>2,5</sup> Since settlement and early waves of migration, the Faroe Islands have been mostly isolated and have experienced minimal immigration and population growth until recent years, with a census size of about 4,000 people in the 1700s<sup>6</sup> increasing to over 54,000 as of August 2023 (<https://hagstova.fo/en/population/population/population>). Patterns of genetic diversity and linkage disequilibrium from few genetic markers reflect the founder event, followed by a historically small population size and subsequent rapid expansion.<sup>7</sup>

Overall, despite the small size and remote location of the Faroe Islands, the above evidence suggests that the genetic makeup of Faroese people may have been influenced by waves of early migration and admixture from various northwest European and Scandinavian populations. However, no population genomic studies have yet been carried out on whole- genome sequencing data to date, with the exception of one study investigating relatedness and autozygosity for a limited sample size of eight individuals.<sup>8</sup> An in-depth analysis of the genomic architecture of the Faroese may reveal how the islands' demographic history has contributed to present-day health and disease in this population. This is of particular interest as the Faroese have a high burden of certain diseases relative to global and other European populations, such as inflammatory bowel disease and type 2 diabetes, among others.<sup>9-18</sup> The Faroe Genome (FarGen) Project set out to understand how the genetic diversity of the Faroese contributes to health.<sup>19</sup>

Moreover, studying the genetic diversity of the Faroese provides potential insight into human migration, adaptation, and population structure in the North Atlantic region. Recently developed haplotype-based methods can provide fine resolution for inferring shared ancestry among individuals and the detection of population-specific haplotypes, but they require panels of whole-genome sequencing data.<sup>20</sup> These methods serve to better account for recent patterns of population structure and cryptic relatedness in population-based genetic studies, particularly in populations like the Faroe Islands with strong founder effects.

In this study, we present the first whole genomes sequenced as part of the FarGen Project. A set of 40 individuals were selected to optimally represent the genetic diversity of the broader Faroese population, with the aid of genealogical data reaching back to approximately 1650 CE. Whole genome sequencing (WGS) data was generated and used to identify putatively functional alleles enriched in the Faroese population, assess ancestry patterns within contemporary genomes, map signals of recent positive selection, and analyze local ancestry in a combined dataset of ancient and contemporary genomes of European descent spanning a period of 3,000 years.

This study provides insight into the genetic variation, demographic history, and selection landscape of the Faroese population. These first whole genomes from FarGen may serve as a useful reference for studies on the broad implications of various evolutionary genetic processes, including bottlenecks, ancient admixture, and positive selection. More in-depth studies may expand this current study by further investigating the genetic architecture of the Faroese to unravel the demographic and evolutionary history of the population and its impact on complex traits and diseases in the islands.

## Results

### Whole Genome Sequencing of Faroese Individuals

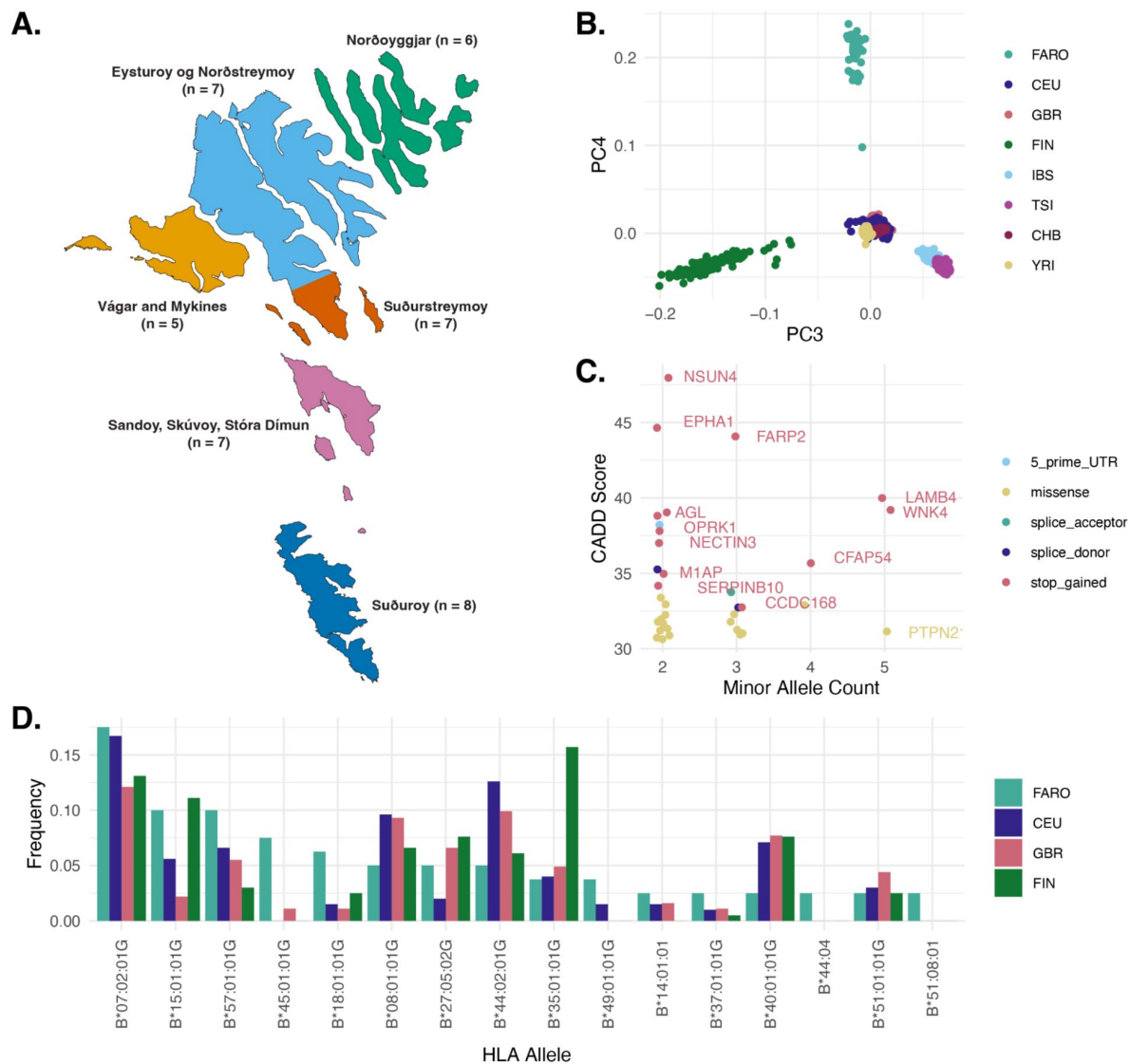
The Faroese Multi Generation Register (<https://fargen.fo/research/multi-generation-registry>) was used to reconstruct a single connected genealogical tree for the 1,541 participants in the first phase of the FarGen cohort.<sup>19,21</sup> Individuals with fewer than six direct ancestors (two parents and four grandparents) recorded in the registry were excluded. Pairwise kinship coefficients were estimated from the genealogical tree and used to perform relatedness pruning with a threshold of  $2^{-6}$  resulting in 332 minimally related individuals. We defined six geographical regions of the Faroe Islands using language dialects, and assigned individuals to these regions based on their place of birth (**Fig. 1A**). A total of 40 minimally related individuals were selected for inclusion in the study, with five to eight individuals sampled from each region.

The genomes of the 40 individuals were sequenced to a median depth of 20x in the Faroe Islands at the FarGen laboratory, and they all passed quality control metrics (**Fig. S1**, **Table S1**). Variant calling was performed jointly with relevant reference genome panels, including 1000 Genomes high-coverage data from Northern European (CEU & GBR, N = 190), Southern European (TSI & IBS, N = 214), Finnish (FIN, N = 99), East Asian (CHB, N = 103), and West African (YRI, N = 108) individuals, and imputation was performed within the cohort using an approach we have previously described.<sup>22,23</sup> The first component of principal component analysis (PCA) on the jointly called genotype data captured the cline between West African (YOR) and all other individuals, and the second component captured the cline between East Asian (CHB) and all other individuals (**Fig. S1**). Principal components three and four separated European individuals, with Faroese individuals forming a distinct cluster from Finnish, Northern European, and Southern European reference groups (**Fig. 1B**).

Variant calls were annotated with predicted functional impact and allele frequency across the reference groups. Using these annotations, we identified 35 putatively functional alleles present in our Faroese panel, that are unobserved in the European mainland reference panels included in this study (CADD > 30 and at least two minor alleles observed in Faroese individuals, and no minor alleles observed in Finnish or Northern European reference individuals, **Table S2**). These included 13 stop gain variants and 18 missense variants, and a maximum minor allele count of 5, corresponding to a frequency of 6.25% in the cohort (**Fig. 1C**).

### HLA-B Allele Frequencies

Observational evidence from the FarGen project found that ankylosing spondylitis (AS) may be at a higher prevalence in the Faroe Islands (unpublished data), however this has not been confirmed by followup epidemiological studies. The major histocompatibility complex (MHC) plays a role in various autoimmune diseases that may be at higher prevalence in the Faroes, including ankylosing spondylitis and other more common diseases like inflammatory bowel disease.<sup>24,25</sup> In particular, HLA-B\*27 is associated with ankylosing spondylitis (AS), with approximately 80-90% of AS patients carrying the HLA-B\*27 allele.<sup>25</sup> It explains about 30% of the heritability, and ~6-8% of European populations are carriers of HLA-B\*27.<sup>25,28</sup> Using the WGS data, we genotyped human leukocyte antigen (HLA) alleles with HLA\*LA.<sup>29</sup> We provide



**Figure 1.**

### A Faroe Islands whole genome reference.

**A)** Map of the Faroe Islands, colored by the six sampling regions. The number of minimally related FarGen participants from each region selected for whole genome sequencing is indicated. **B)** Principal component analysis (PCA) of Faroe Islands genomes jointly called with relevant 1000 Genomes reference data shows separation of European groups by PCs 3 and 4 (FARO, Faroe Islands, CEU, Central Europeans, GBR, British, FIN, Finnish IBS, Iberian, TSI, Tuscan, CHB, Han Chinese, YRI, Yoruban). **C)** Faroe Islands enriched putatively functional alleles visualized by minor allele count, CADD score, and Variant Effect Predictor (VEP) consequence. Variants shown are those with CADD > 30 and at least two minor alleles observed in Faroe Islands individuals, and no minor alleles observed in Finnish or Northern European reference individuals. **D)** HLA-B allele frequencies for alleles detected at least twice in Faroe Islands individuals. In this cohort, 1 minor allele corresponds to an allele frequency of 1.25%.

HLA-B allele counts and allele frequencies in the Faroese cohort as well as the allele frequencies in 1000 Genomes British (GBR), Central European (CEU), and Finnish (FIN) individuals (Table S3). To the best of our knowledge, there have not been any larger studies of HLA-B allele frequencies in Faroese individuals, and none are currently recorded in the Allele Frequency Net Database.

The most frequent HLA-B allele in the Faroese cohort is HLA-B\*07:02 (17.5% of haplotypes), which is a common haplotype in European populations (Fig. 1D). We did not observe a substantial difference in HLA-B\*27 allele frequency in Faroese individuals (6.25% with four B\*27:05 and one B\*27:01 calls) as compared to other European reference groups (2-7.6%). While 80-90% of people of European ancestry with AS carry the HLA-B\*27 allele, only ~6% of HLA-B\*27 carriers in the US and Europe have AS. The low frequency of the HLA-B\*27 allele in this Faroese cohort in the population broadly suggests that if AS is at a higher prevalence, there may be other underlying genetic or environmental factors that explain some of the increased risk.

## Population Structure and Relatedness

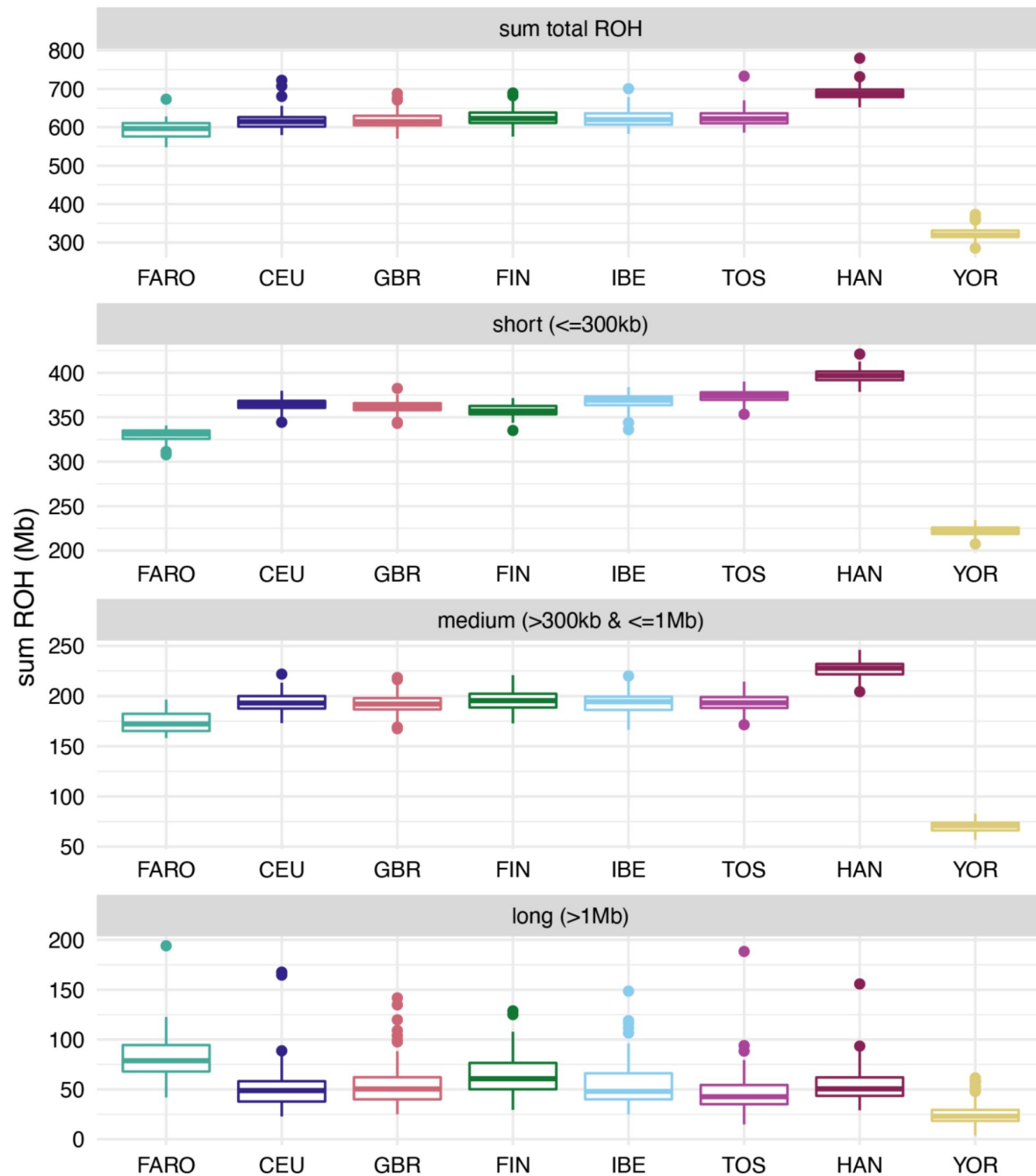
Pairwise kinship was calculated in the cohort using *popkin*. Clustering by geography was observed when including global reference populations in the kinship calculation, however the Faroese individuals do not show obvious clustering by region, which is expected given the relatedness pruning during sample selection (Fig. S2A-B).

We also looked at runs of homozygosity (ROHs) in the Faroese and reference cohorts (Fig. 2), which can provide insights into the demographic history of the population. As the Faroese population likely experienced a founder event during the settlement of the islands followed by rapid population size expansion in recent generations, we would expect to see more of the genome contained in ROHs compared to other global populations that have not experienced as strong a bottleneck. When looking at the sum total amount of the genome in ROHs, we found overall elevated levels of ROH in the European and Asian groups included in this analysis, most likely reflecting ancestral out-of-Africa bottlenecks for Eurasian populations (Fig. 2, top panel). However, the Faroese population did not have an elevated total amount of the genome contained in ROHs compared to other European groups.

To explore this further, we calculated the sum total amounts of ROH at different size categories (small, medium, large) (Fig. 2, bottom 3 panels). We see that, on average, the Faroese individuals have less of the genome contained in short (<300 Mb) and medium (> 300 Kb and <= 1 Mb) ROHs compared to other European groups, but more of the genome contained in long (> 1Mb) ROHs. Short and medium ROH are chunks inherited from older ancestors and reflect older events, for example, an ancient population bottleneck or founder event that has resulted in lower overall haplotype diversity, yet with enough time for recombination to break up haplotypes, while long ROH reflects chunks inherited from recent ancestors and can reflect more recent bottleneck events. Interestingly, the average amount of an individual's genome that is contained in ROHs extending over 1Mb in length is higher in the Faroese population (~82.5 Mb) than the Finnish reference individuals (~63.9 Mb) and any of the other groups analyzed, which is consistent with a more recent or stronger bottleneck or founder event.

## Signals of Positive Selection

We investigated signals of recent or ongoing positive selection in this Faroese cohort by calculating both the integrated haplotype score (iHS) using *hapbin*. The sample size of the WGS cohort is relatively small (n=40), so our ability to detect signals of selection is limited. For comparison, we also calculated iHS in British individuals from the 1000 Genomes (GBR) that were included in joint calling and passed subject-level filters (n=90). The iHS values were standardized genome-wide and two-tailed p-values were computed according to the standard normal distribution. We additionally calculated q-values (the minimum False Discovery Rate (FDR) should a test be considered significant) for each test, and determined p-value significance thresholds at which FDR < 0.01



**Figure 2.**

### Runs of homozygosity by group.

Amount of the genome (Mb) contained in runs of homozygosity (ROH) stratified by group. Top panel is the sum total of the genome contained within ROH, with the other panels showing this split by length (short, medium, and long).



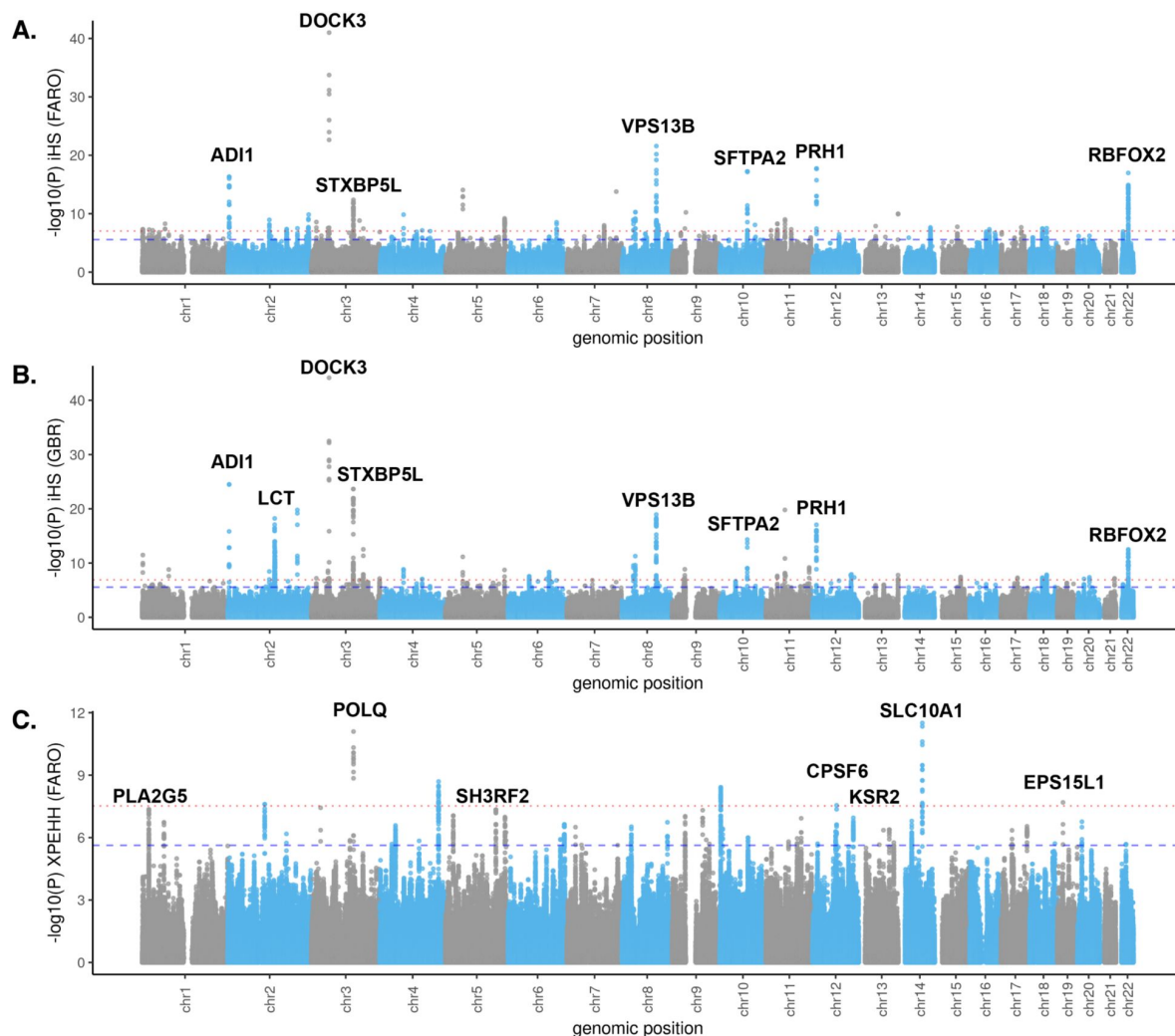
and  $FDR < 0.001$  for each population (see Methods). We observed that a number of significant selection signals were shared between the Faroese and British cohorts, which is not unexpected given the relationship between these two populations (**Fig. 3A-B**). The strength of these signals did differ from one population to the other, though this may be due to differences in sample size or changes in selection pressure after the populations diverged. To better identify population-specific signals, we also calculated the cross-population expected haplotype homozygosity (XPEHH) comparing Faroese and British haplotypes, and identified significance cutoffs following the same approach described above (**Fig. 3C**).<sup>43</sup> Any extreme positive values of this statistic indicate longer haplotypes at a focal marker in the Faroese cohort compared to the British cohort, while extreme negative values indicate the reverse. Therefore, positive values are indicative of selection signals in the Faroese cohort. Across both tests, we highlight 20 loci with the most extreme values for these statistics, serving as evidence of positive selection in the Faroese genomes at those loci (**Table S4**).

One signal that has been consistently observed across northern European populations is in the *LCT/MCM6* region, corresponding to positive selection for lactase persistence alleles.<sup>40,43,45</sup> Interestingly, this region showed strong iHS signals ( $|\text{standardized iHS}| > 8$ ) in GBR and was considered genome-wide significant in our analysis (minimum  $p = 5.95 \times 10^{-19}$ ,  $q = 7.82 \times 10^{-14}$ ) (**Fig. 3B**), but the signal is weaker in the Faroese ( $|\text{standardized iHS}| > 4$ ) and was not considered significant in our analysis (minimum  $p = 3.57 \times 10^{-6}$ ,  $q = 0.0118$ ) (**Fig. 3A**). To investigate the haplotype structure further, we plotted the decay in expected haplotype homozygosity (EHH)<sup>46</sup> and haplotype furcation around one of the lactase persistence alleles (rs4988235; chr2\_135851076\_G\_A) for the Faroese and British cohorts using the *rehh* package (**Fig. 4A-D**).<sup>47</sup> The decay and furcation plots are centered around the focal marker, and a furcation occurs when unique haplotypes arise at an allele, similar to a tree splitting into branches. Thicker branches in the furcation plot indicate higher frequency of that haplotype in the population. Significant differences between the furcation patterns for an “ancestral” (reference) and “derived” (alternate) allele correspond to extreme iHS values and therefore are indicative of strong positive selection. For an alternative view of the region, we used *Haplostrips* to visualize the haplotype structure from chr2:135677850-135986443 (**Fig. 4E**).<sup>48</sup> From these plots, we observed far less diversity on the lactase persistence haplotype in GBR, consistent with a stronger selection signal. This may be explained by shared selection on the ancestral northern European branch followed by either relaxed selection for lactase persistence or population-specific drift in the Faroes after the population split from other northern European groups and settled the archipelago.

One of the top XPEHH signals in the Faroese WGS cohort included variants in *SLC10A1* (**Fig. S3**), a sodium/bile acid transporter that plays a role in circulating bile salts to and from the liver and small intestine for the absorption of dietary fat and fat-soluble vitamins such as vitamin D.<sup>49–52</sup> *SLC10A1* deficiency has been associated with familial hypercholanemia, or elevated concentrations of bile acids, which can lead to fat malabsorption and vitamin D deficiencies among other secondary health conditions.<sup>50,53,54</sup> Another top XPEHH signal included variants in *POLQ*, encoding for a DNA polymerase which plays a role in DNA repair (**Fig. S4**).<sup>55–59</sup> *POLQ* has been shown to be involved in various cancers in mice and humans, in particular skin, stomach, lung, breast, and colon cancers.<sup>55,59–62</sup>

## Fine-Scale Structure and Connections to Ancient Genomes

Given that early Faroese settlers have documented historical relations to both Northern European Vikings and Northwestern European Celtic communities, we sought to study fine-scale genome-wide ancestry relationships between the sequenced Faroese genomes and publicly available ancient genomes from Iron Age and Viking Age Europe. We downloaded 616 ancient imputed genomes from Allentoft *et al.* 2024, spanning from the present-day to the late Bronze Age from Europe and focusing specifically on West- and North-Europe, including ancient Faroese

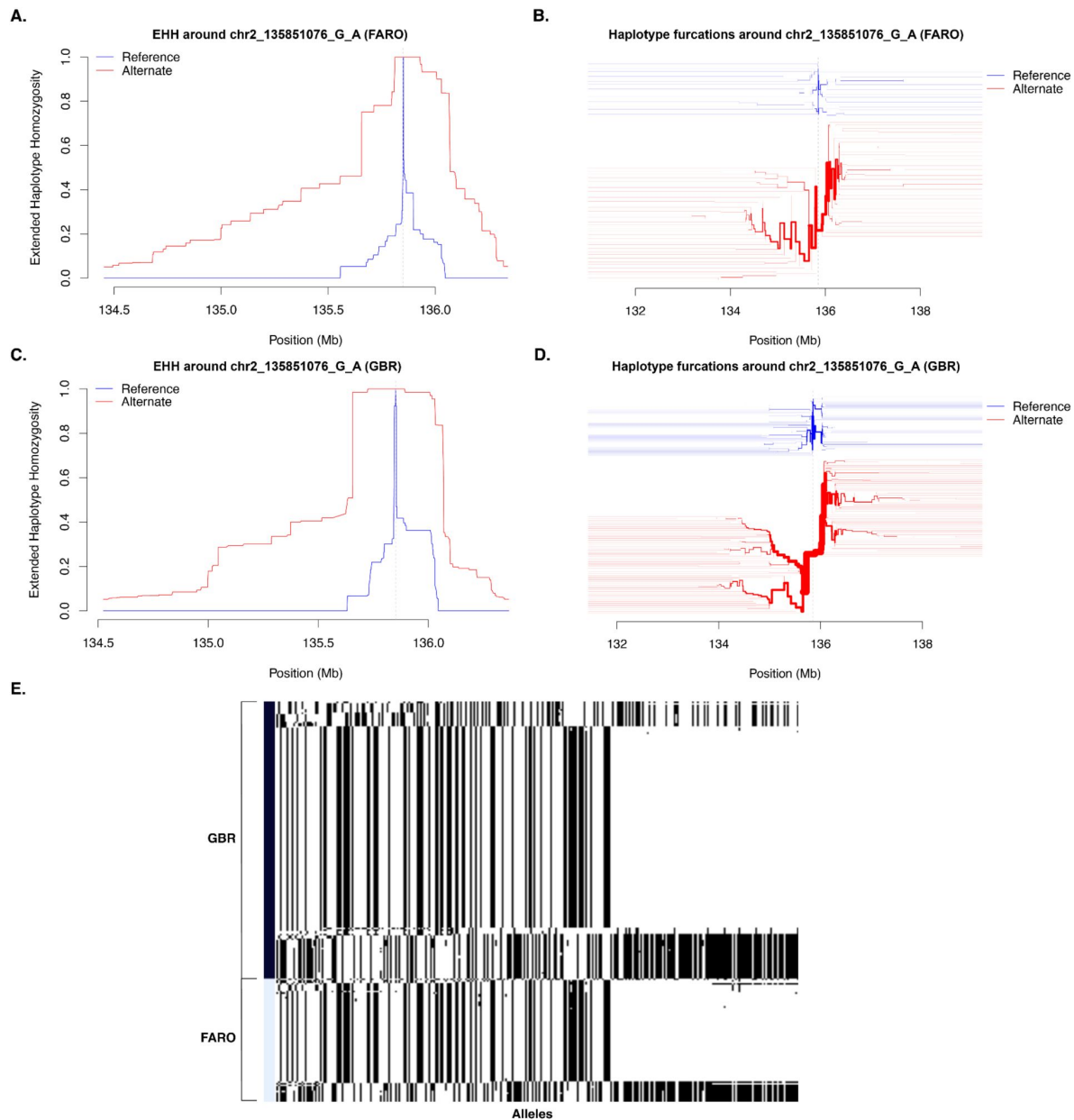


**Figure 3.**

### Selection scan results for Faroese and British cohorts.

**A)** Log transformed two-tailed p-value of the standardized integrated haplotype score (iHS) in the 40 Faroese genomes (FARO). **B)** Log transformed two-tailed p-value of the standardized iHS for 90 British WGS samples from 1000 Genomes (GBR). **C)** log transformed two-tailed p-value for the standardized cross-population expected haplotype homozygosity (XPEHH) for FARO vs GBR (only positive values, which indicate selection in FARO, are plotted). Some genes in the top loci are indicated on each plot. The p-value cutoffs which correspond to a False Discovery Rate (FDR) at 0.01 and 0.001 are respectively indicated by the red dotted line and blue dashed line in each plot. **A)** For iHS in FARO, these cutoffs are  $p = 2.72 \times 10^{-6}$  (FDR = 0.01) and  $p = 9.20 \times 10^{-8}$  (FDR = 0.001). **B)** For iHS in GBR, the cutoffs are  $p = 2.78 \times 10^{-6}$  (FDR = 0.01) and  $p = 1.75 \times 10^{-7}$  (FDR = 0.001). **C)** For XPEHH in FARO vs GBR, the cutoffs are  $p = 2.35 \times 10^{-6}$  (FDR = 0.01) and  $p = 3.01 \times 10^{-8}$  (FDR = 0.001). See Methods for details on p-value and FDR estimation.





**Figure 4.**

#### Haplotype visualizations for the LCT/MCM6 locus.

**A)** Decay in Expected Haplotype Homozygosity (EHH) and **B)** haplotype furcation plot for FARO centered on lactase persistence allele rs4988235; chr2\_135851076\_G\_A. **C)** Decay in EHH for GBR and **D)** haplotype furcation for GBR centered on the same allele. **E)** *Haplotrips* visualization of haplotype structure in the region chr2:135677850-135986443. In this panel, columns correspond to segregating alleles, and rows correspond to individuals. In the haplotype furcation plots (panels **B** & **D**), the haplotypes for the reference allele (G) are in blue, and those for the alternate allele (A) are in red.

genomes.<sup>63</sup> We incorporated these genomes into a combined panel including our present-day Faroese dataset and used the software HaploNet to infer fine-scale population structure based on patterns of haplotype similarity across the genome.<sup>20</sup>

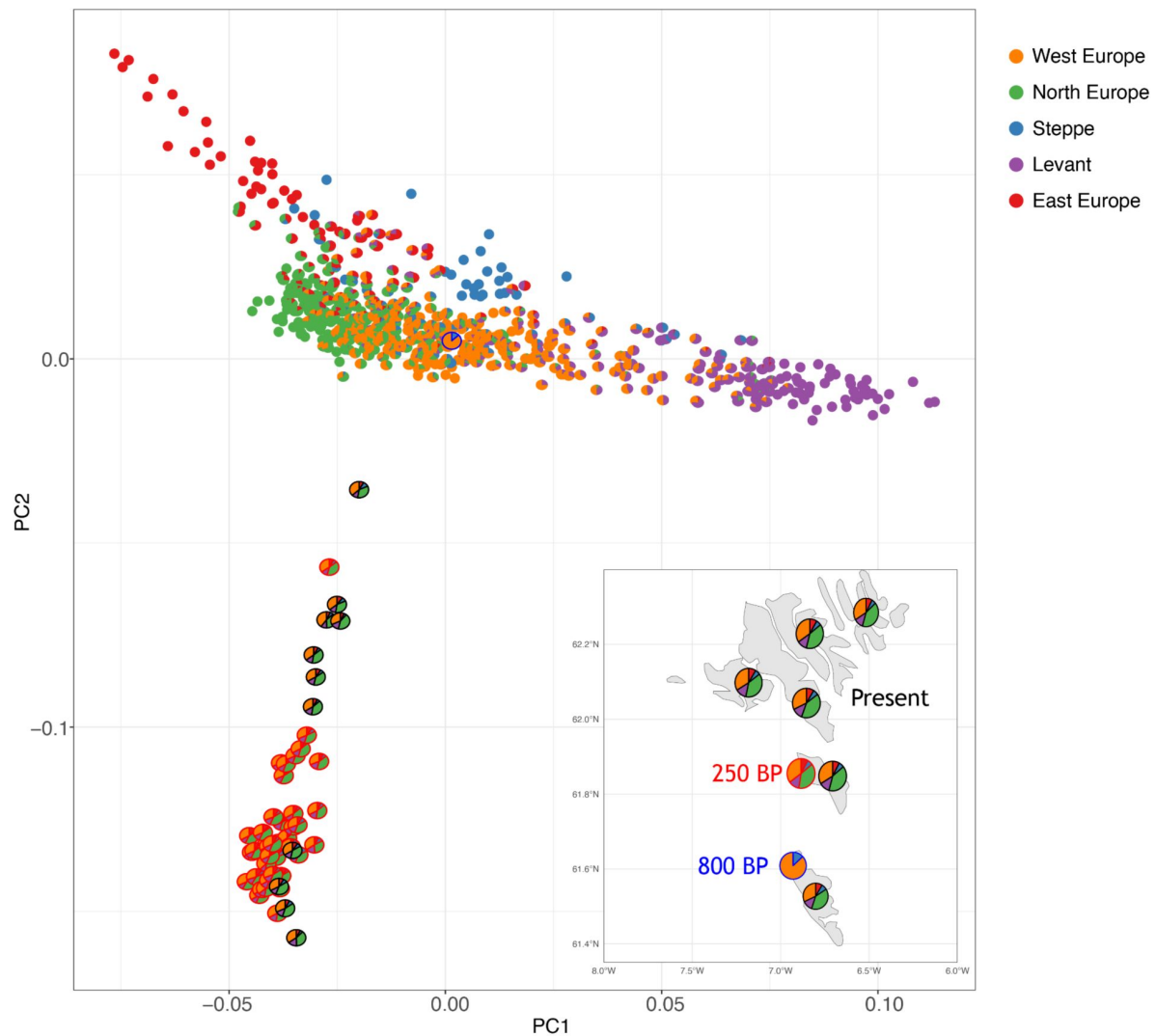
HaploNet identified five ancient sources, through unsupervised ancestry estimation. We then used HaploNet to model both ancient and present-day Faroese genomes as composites of any of the five ancestries through supervised ancestry estimation. We used the ancestral haplotype cluster frequency estimates from individuals not found in the Faroe Islands in order to estimate the admixture proportions of these sources in the European mainland. We used these admixture proportions to label the sources based on the locations in the map where these ancestries tend to be maximized in the Iron Age and Viking Age periods. The resulting labels were: “Steppe”, “East Europe”, “Levant and East Mediterranean”, “West Europe” and “North Europe”. For example, the ancestry labeled “West Europe” is maximized in individuals predominantly found in Celtic contexts (e.g. Roman and Iron Age Britain, Iron Age France) while the ancestry “North Europe” is maximized in individuals characteristic of historically Viking or pre-Viking contexts (e.g. Iron Age individuals from Denmark, as well as Viking Age individuals from Denmark, Norway, Sweden, and Estonia). However, we note there is no one-to-one correspondence between archaeological context and genetically inferred ancestry, and that many mainland individuals contain inferred ancestries from diverse sources. Indeed, Margaryan *et al.* 2020 showed that Viking-context individuals can derive ancestries from multiple Bronze and Iron Age sources across Europe.<sup>64</sup>

We then focused on the frequency of the ancestry sources in the Faroese individuals. We find that the present-day Faroese individuals are predominantly composed of roughly equal proportions of “West” and “North Europe” ancestry, while “East” and “South Europe” ancestries are detected at much lower frequencies. Present-day ancestry proportions are identical to those found in the Faroese ancient samples from the Sandur church site in Sandoy and dated to the 17th or 18th centuries based on their archaeological contexts (Fig. S5).<sup>64</sup> Margaryan *et al.* also sequenced a Faroese sample that was excavated from the á Bønhúsfløtu site in the village of Hvalbøur on Suðuroy, and was contextually-dated to be approximately 800 years old. This individual is inferred to be almost entirely composed of “West Europe” ancestry (Fig. S6).

We utilized haplotype cluster likelihoods to explore population structure in the same set of samples. When plotting ancient European samples together with the Faroese samples (Fig. 5), we observe that present-day Faroese individuals (circled in black) separate from the ancient Europeans along the second principal component (PC2), as do the older Faroese samples from the 17th and 18th centuries (circled in red). This perhaps suggests a bottleneck process that differentiates the 17th/18th-century and present-day Faroese from the rest of the ancient European samples, in concordance with the above ROH results. Notably, the 800-year-old sample of a Faroese individual with predominantly “West Europe” ancestry does not fall along the Faroese PC2 cline, suggesting that this individual might predate the bottleneck.

## Discussion

Here we presented the first whole-genome sequence data from 40 minimally related individuals from across the Faroe Islands, a North Atlantic founder population. The Faroese have a high prevalence of several diseases in comparison to other European or global populations, several of which have been of particular interest in epidemiological studies of the region (e.g. inflammatory bowel disease, type 2 diabetes, multiple sclerosis, and ankylosing spondylitis).<sup>9–18</sup> We investigated the frequency of HLA-B27 in the Faroe Islands, an allele which has been previously associated with ankylosing spondylitis (AS). We found that despite the observed high prevalence of AS in the Faroe Islands there was no evidence of increased HLA-B27 allele frequency compared to other European populations. However, a stop gain variant in *SERPINB10* was among those enriched in the Faroese cohort, and it may contribute to increased AS risk (rs138084090, AF =



**Figure 5.**

**PCA analysis of 616 ancient imputed genomes from Europe and 40 present-day Faroese genomes.**

Each individual is depicted as a pie chart, showing ancestry proportions estimated using Haplonet. Ancestry proportions for ancient individuals were estimated unsupervised, while those for present-day Faroese individuals were estimated semi-supervised using ancient genomes as references. The five colors represent different ancestral sources: orange for West Europe, green for North Europe, blue for Steppe, purple for the Levant and East Mediterranean, and red for East Europe. The geographical distribution (bottom-right) highlights historical samples (250 years BP) in red, this study's samples in black, and an 800-year-old individual sample in blue.

2.5%). Rare variant analysis of this gene in the UK Biobank found that an independent stop gain variant in the same gene is nominally associated with increased AS risk (rs145346731,  $p = 2.5 \times 10^{-4}$ ), which was the most significant phenotypic association for the gene.<sup>65</sup> *SERPINB10* is most highly expressed in neutrophilic metamyelocytes,<sup>66</sup> a cell type with disease relevance to AS.<sup>67</sup> *CCDC168*, another gene with a Faroese enriched stop gain variant had highly significant rare variant associations in the UK Biobank with corneal hysteresis and intraocular pressure (rs1361247423,  $p = 1.79 \times 10^{-57}$ ,  $p = 5.21 \times 10^{-13}$ , respectively). Finally, a Faroese enriched stop gain variant in *AGL* identified in this study (rs113994128) has been previously reported as causing glycogen storage disease type IIIA in the Faroe Islands, which is estimated to have the highest prevalence of the disease world-wide.<sup>68</sup> These findings emphasize the importance of further research into the role that unique genetic variation in the Faroe Islands may play in the incidence of diseases that, while common in the Faroes, have a global burden as well.

We found elevated amounts of the genome contained in long runs of homozygosity (ROHs) compared to other European reference cohorts, including another founder population from Finland. The higher proportion of the Faroese genome contained in these long ROHs suggests a stronger or more recent bottleneck in the Faroese population history. With the second phase of the FarGen study, a larger sample size will facilitate investigation into the timing and severity of the bottleneck(s). Additionally, founder populations such as the Finnish, have been a common focus for studies of founder events, genetic isolation, and the effects of haplotype sharing on aspects of human health and disease.<sup>69–72</sup> As such, the Faroese population may serve as another useful global reference for studying the influence of demographic history on genetic variation and trait architecture. Long ROHs, in particular, can be enriched for deleterious variation or be of interest in understanding the genetic architecture of health-related traits in human populations.<sup>39,73</sup> Studying these long ROHs may be relevant for future studies of health outcomes or other traits of interest in the Faroese population.

We also detected several regions under recent positive selection in the population. We found that many of the top selection signals were shared between the Faroese and British cohorts, which is unsurprising given the recent divergence between these two populations. We did find differences in the strength of these signals; for example, there is more diversity on the Faroese *LCT/MCM6* lactase persistence haplotypes. The lactase persistence allele rs4988235 (chr2\_135851076\_G\_A) has been inferred to be under strong selection at least until the medieval period in northwestern European groups.<sup>74,75</sup> Patterson et al. 2021 found that the rate of increase in allele frequency may have slowed in recent periods; however, this does not exclude the possibility of continued or fluctuating selective pressure as this is consistent with the expected sigmoidal trajectory for an allele under ongoing selection.<sup>76</sup> While the increased diversity on the Faroese lactase persistence haplotypes may be simply explained by population-specific drift, this result could also indicate relaxed selection for lactase persistence alleles after settlement of the Faroe Islands, possibly due to changes in dietary habits in the new environment. The traditional diet of the Faroe Islands consisted of a higher reliance on animal and marine fats such as sheep tallow, whale blubber, and liver from codfishes, while dairy products such as milk and cheese, and particularly that from cattle, were more limited in availability.<sup>77</sup> Based on the ancient genomes for which we have available data, we can attest that the selected variant at rs4988235 was present at high frequencies already in the 17th/18th centuries, and imputation further suggests the haplotype containing the allele to at least have been present in the islands 800 years ago.

We detected selection targets that were specific to the Faroese population using the XP-EHH statistic with the British cohort as the comparison population. One top selection signal is in *POLQ*, which plays a role in DNA repair and various cancers. Without collecting relevant phenotypes or environmental factors, it is difficult to hypothesize what selection pressure may be driving the strong signal in *POLQ*, so this will be an important area of follow-up for future studies. In another top signal, we find *SLC10A1* which plays a role in fat and vitamin D absorption. Positive selection related to differences in dietary fat intake has been hypothesized in many human populations,

such as the Inuit population in Greenland.<sup>78,79</sup> Also situated in a far northern latitude, the Faroese diet is similar to that of the Inuit population, relying on animal and marine fats.<sup>77,80</sup> The relationship between *SLC10A1* and vitamin D levels may also be relevant, as the northern latitudes of the Faroe Islands and minimal UV exposure can lead to vitamin D deficiency, which has been hypothesized to be a strong selection pressure in populations in extreme latitudes.<sup>81,82</sup>

Although we hypothesize that these results suggest possible adaptations to environmental pressures of diet or UV exposure in northern latitudes, we cannot draw definitive conclusions based on this current study. It is certainly possible that variants that have risen to high frequencies due to past or ongoing positive selection now play a role in health outcomes in modern populations. For example, the gene *TBC1D4*, which was shown to be under positive selection in the Greenlandic Inuit population likely due to a historical diet low in carbohydrates, has been associated with type 2 diabetes and insulin resistance in the same population.<sup>80,83</sup> The prevalences of some diseases enriched in the Faroese population may be related to genomic regions under positive selection. For example, the results of several studies have suggested a role of vitamin D deficiency in the development of multiple sclerosis.<sup>84–86</sup> Future studies could involve collection of relevant phenotypes and focus on characterizing selective pressures and fine-mapping targets of selection as have been done in studies that more thoroughly characterized selection signals related to dietary adaptation and UV exposure and their functional consequences in other northern latitude populations.<sup>79,81,83</sup>

We have inferred ancestry tracts in the present day Faroese genomes that were inherited from ancient populations throughout Europe. We found that present-day Faroese individuals have similar relative ancestry contributions from past “North” and “West Europe” Iron Age populations. The most ancient genome available from the Faroe Islands matches ancestry patterns found in Iron Age Wet Europe. This suggests that the “North Europe” ancestry may have arrived in the islands afterwards. This could have occurred either via a mixture of the original “West Europe” ancestry with individuals of predominantly “North Europe” ancestry, or a by replacement with individuals that were already of mixed ancestry at the time of arrival in the islands (the latter are not uncommon in Viking Age mainland Europe). Our analysis also suggests a bottleneck or a more progressive differentiation process in the islands relative to the mainland, which may postdate the most ancient Faroese genome currently available (approximately 800 years old). The most ancient Faroese sample from Margaryan *et al.* - composed almost entirely of “West Europe” ancestry - is a male individual found in a chapel-site in Suðuroy. Consistent with this, a local legend suggests this site may have been occupied by Irish monks.<sup>87</sup> More ancient and present-day samples from the islands could shed further light on the history of the Faroese population.

This study focused on population genomic analyses such as selection scans, population structure, kinship, and ancestry. Given the unique settlement history and genetic architecture of the Faroe Islands, future studies which combine genomic data with relevant phenotype data could provide useful insight into the underlying genetic mechanisms of those traits. In particular, larger-scale genomic studies in the Faroese could investigate genetic risk factors which contribute to the high prevalence of autoimmune and metabolic disease on the islands. This is a focus of the second phase of the FarGen study, which is currently ongoing.

## Methods

### Sample Selection and Cryptic Relatedness

## FarGen cohort

The participants in this study voluntarily enrolled in the FarGen project (The Faroe Genome Project: <https://www.fargen.fo/en/home/>). The 1,541 subjects are extensively described in Apol *et al.* 2022.<sup>21</sup> Participant inclusion criteria for the FarGen project are that participants must live in the Faroe Islands or be of Faroese descent. Apol *et al.* report that 96.4% of the participants have between one and four Faroese grandparents. The cohort has a mixed health status composition, with 75% of the participants self-reporting that they have a confirmed diagnosis. Apol *et al.* found that the cohort is somewhat biased in terms of geographical representation, with the capital region being substantially over-represented.

## Reconstruction of genealogy

The Multi-Generation Register at the Faroese Health Authority describes the ancestry of inhabitants of the Faroe Islands (<http://fargen.fo/research/multi-generation-registry>). The lineages can be traced back to approximately 1650 C.E. The register records birth date, parent identities, parents' residence at the time of birth, and more. The Legacy Family Tree (<https://legacyfamilytree.com>) genealogy software is used to manage the digitized records. We reconstructed a genealogical tree of all the individuals in the FarGen cohort by looking up each individual, and recursively looking up their parents until there are no more ancestors. After reconstructing the genealogy of each individual two generations in the past, we discarded any who had fewer than 6 direct ancestors recorded in the Multi-Generation Registry (2 parents and 4 grandparents). We note that although an individual is recorded in the register, there is no guarantee they were born in the Faroe Islands.

## Geographical stratification through dialect

We defined six geographical regions of the Faroe Islands as annotated in **Fig. 1A**: Norðoyggjar; Eysturoy and Norðstreymoy; Suðurstreymoy; Vágur and Mykines; Sandoy, Skúvoy, and Stóra Dímun; Suðuroy, and placed individuals within these regions based on birth place. The boundaries of the regions were defined using isoglosses (i.e. boundaries where we see changes in dialect) as described in Bráinsson 2012.<sup>88</sup> For example, the isogloss for "á", which may be pronounced either as [a:] or [ɔa], separates Norðoyggjar in the north from the rest of the islands.

## Calculating pairwise kinship coefficients

In order to avoid sequencing highly related samples, we used the large constructed genealogy to account for cryptic relatedness. We calculated pairwise kinship coefficient between every individual using *kinship2*.<sup>89</sup> This method assumes that the founders (individuals in the pedigree without recorded parents) are unrelated to other founders and each individual founder's parents were not related to each other, which may not always be the case in this population.

## Sample selection using graph theory

We constructed a relationship graph with nodes representing individuals, and connected two nodes by an edge if their kinship coefficient is above a given threshold as described below. Ideally, we would remove nodes such that all edges are removed, while keeping as many nodes as possible, referred to as the *maximum independent set* problem. Obtaining an exact solution to the maximum independent set is an NP-hard problem (exponential time complexity), making it infeasible for our applications. Instead, we obtained an approximate maximum independent set using an algorithm described in Boppana *et al.* 1990.<sup>90</sup> We performed relatedness pruning with a threshold of  $2^{-6}$  on 1,294 FarGen individuals who were not missing a birth region, resulting in



332 individuals. This was the minimum threshold for which we could have enough sampling from each region. From these 332 individuals, we sampled 5 to 8 from each region, yielding 40 individuals for whole genome sequencing and subsequent analyses.

## Bioinformatics

### Sequencing and quality control

DNA samples from the 40 selected individuals were sequenced at FarGen (using TruSeq PCR-free libraries on Illumina NextSeq 500 instruments) to an average depth of 19.2x, ranging from 9.9x to 31.5x per sample. Sequencing QC was investigated with FASTQC, Picard (including CollectWgsMetrics for coverage) and VerifyBamID for contamination.

### Variant calling and imputation

Reads were processed with Variant Bio's in-house processing pipeline based on the GATK Best Practices (CCDG functional equivalence version).<sup>91</sup> Joint genotyping was performed with GATK (version 4.2.0.0) including 714 genomes from the 1000 Genomes Project (503 Europeans, 103 Han Chinese, 108 Yoruba) and followed by VQSR (`--truth-sensitivity-filter-level` to 99.8 for SNPs and 99.0 for indels). Only PASS variants in GIAB high-confidence regions (~80% of GRCh38) were retained.<sup>92</sup> Genotypes with  $GQ \leq 20$  were filtered (set to missing) and imputation was performed within the full cohort of 754 genomes using Beagle v5.1. Variants enriched in the Faroese cohort and with predicted functional impact (Table S2) were additionally hard-filtered with  $VQSLOD > 20$ .

### HLA typing

We ran HLA\*LA (v1.0.3) on the mapped reads to determine HLA types for the 40 Faroese individuals as well as for the GBR, CEU, and FIN reference population individuals from 1000 Genomes.<sup>29</sup> Benchmarking the method on 1000 Genomes Project data, where HLA types are known, we estimate overall HLA-B typing accuracy at 93.4%, 88.8% and 89.9% for the GBR (N=91), CEU (N=99), and FIN (N=99) reference populations. Accuracy of B27 detection specifically is 100% (N=10), 83.3% (N=6) and 100% (N=15) based on these three reference cohorts, respectively, with one B\*27:05 allele mis-identified as B\*27:26 in CEU. Table S3 contains counts, mean quality scores, and frequencies of all HLA-B alleles detected among the 40 Faroese as well as the GBR, CEU, and FIN reference populations.

## Population Genetics

### Individual and variant-level filtering

Beginning with a total of 21,837,577 variants and 754 individuals, we applied various individual-level and variant-level quality control filters for downstream analyses. We filtered 8 individuals with mismatched sex based on genetics and reported information. We additionally filtered 5 individuals that were, for any of PCs 1-10, further than 7 standard deviations from the mean. We then filtered 481,403 variants that were not in GIAB high confidence regions, had MSQ rank sum not equal to zero or failed gnomAD v3 QC.<sup>93</sup> We removed 3,627 variants with a minor allele count of less than 1 after individual-level filters were applied. We removed 695,947 variants that were not autosomal. This final dataset included 20,656,600 variants and 741 individuals.

## Selection scans

iHS and XP-EHH were calculated using the *hapbin* software (<https://github.com/evotools/hapbin>) with option `--max-extend 1000000` and `--minmaf 0.05`.<sup>41</sup> All other options were set to default. To compute P-values, we used the method by Fariello et al. (2013), exploiting the fact that detectable regions under strong selection affect a small portion of the genome.<sup>94</sup> For both statistics, values were standardized genome-wide in 2% allele frequency bins, as allele frequency is correlated with allele age and therefore haplotype length.<sup>40,43</sup> We first computed outlier-robust mean and standard deviation with the `rlm()` function from the MASS package in R, to reduce the influence of outliers.<sup>94,95</sup> The standardized values of these summary statistics represent z-scores. We calculated two tailed p-values using these z-scores, giving the probability that we observe these data by chance compared to null expectations for the standard normal distribution. Q-Q plots and histograms of p-values for each statistic are provided in the Supplementary Materials (Fig. S7). For each summary statistic distribution, we also calculated the p-value cutoffs that correspond to a False Discovery Rate (FDR) of less than 0.01 and 0.001, using the q-value R package (<https://github.com/StoreyLab/qvalue>).<sup>42</sup>

EHH decay plots and haplotype furcations for the *LCT* locus were calculated using the *rehh* R package (<https://cran.r-project.org/web/packages/rehh/index.html>)<sup>47</sup> and visualized the haplotype structure of the genomic region chr2:135701076-136009184 using *haplostrips* and plot option `-S 3` (<https://bitbucket.org/dmarnetto/haplostrips>).<sup>48</sup>

## Kinship and runs of homozygosity

The kinship matrix in the WGS cohort was calculated using the *popkin* software (<https://github.com/StoreyLab/popkin>).<sup>36–38</sup> We restricted the analyses to biallelic SNPs with a minor allele frequency of at least 0.01 in at least one subpopulation (i.e. YRI, CHB, FIN, CEU, GBR, TSI, IBS, or FARO), resulting in a dataset of 15,206,409 variants. When calculating the kinship matrix for the Faroese WGS cohort only, we used the `rescale_kinship()` function, which will change the most recent common ancestor and give different absolute values, but the overall relationship structure in the subpopulation remains the same. Using the same data set, we calculated runs of homozygosity (ROHs) for each individual using *bcftools/RoH*.<sup>96</sup> “Short” ROHs were classified as ROH less than or equal to 300 kb, “medium” as greater than 300kb and less than or equal to 1 Mb, and “long” as greater than 1 Mb.

## Fine-scale structure estimation using ancient genomes

A panel of 616 imputed ancient genomes from Allentoft et al. 2024 (downloaded from <https://doi.org/10.17894/ucph.d71a6a5a-8107-4fd9-9440-bdafdfe81455>), representing individuals from several European regions (southern Europe, western Europe, northern Europe, eastern Europe, and central Europe) was used for analyses.<sup>63</sup> Only samples that were estimated to be no older than 3,000 years old were used. Out of these ancient samples, 11 were excavated in the Faroe Islands, 10 of them are historical samples dated to approximately 250 years old, and 1 of them is dated to be approximately 800 years old.<sup>64</sup> The sample location, approximate age, and sources for these samples are listed in Table S5. To consolidate the two panels, we first performed a liftover of the ancient genome VCF files to the GRCh38 reference genome. Following this, we applied quality filters to the dataset (bi-allelic sites MAF > 0.05 and imputation INFO >= 0.5).

We used HaploNet - a neural network-based method for performing window-based haplotype clustering across the genome - for fine-scale population structure inference on the combined panel. HaploNet uses a hidden Markov model to find an optimal window-based local ancestry “painting” across a genome, given estimated haplotype cluster likelihoods, haplotype cluster frequencies, and global ancestry proportions.<sup>97</sup> The Faroese panel’s haplotype frequencies are very homogeneous and highly differentiated from mainland Europeans. For this reason, under an

initial round of unsupervised ancestry estimation, we found that the Faroese individuals captured a major component at first split ( $K=2$ ). We therefore implemented and utilized a semi-supervised ancestry estimation feature in HaploNet<sup>20</sup>. We performed haplotype clustering in non-overlapping windows of 512 SNPs, and we used the resulting haplotype cluster likelihoods to perform principal component analysis (PCA) and estimate both global and local ancestry in the Faroese individuals.

We performed global ancestry estimation in HaploNet using its EM algorithm to find the maximum likelihood estimates using only ancient European individuals (excluding the Faroese individuals)<sup>20</sup>. We used the EM algorithm a second time to estimate the ancestry proportions (Q matrix) in the Faroese individuals. The estimated haplotype cluster frequencies (F matrix) were kept fixed, which means that the semi-supervised approach can be seen as modeling the Faroese individuals using inferred haplotype clusters from the ancient European individuals.

## Data availability

Variant-level summary statistics and genome-wide selection scan results for iHS and XPEHH are available upon request for research that is in line with informed consent and ethical approval. Genetic and meta data from this study is stored at the Faroese Health Authority.

Access to individual-level data is available for research upon participants' re-consent. Researchers will be granted access to de-identified genetic data and metadata, provided that the project protocol has been approved by the Faroese Scientific Ethical Committee and a template material/data transfer agreement has been signed with the Faroese Health Authority in compliance with GDPR (see Gregersen *et al.*, 2021). Requests should be made to Noomi O. Gregersen (noomi@fargen.fo).

## Supplementary Figures

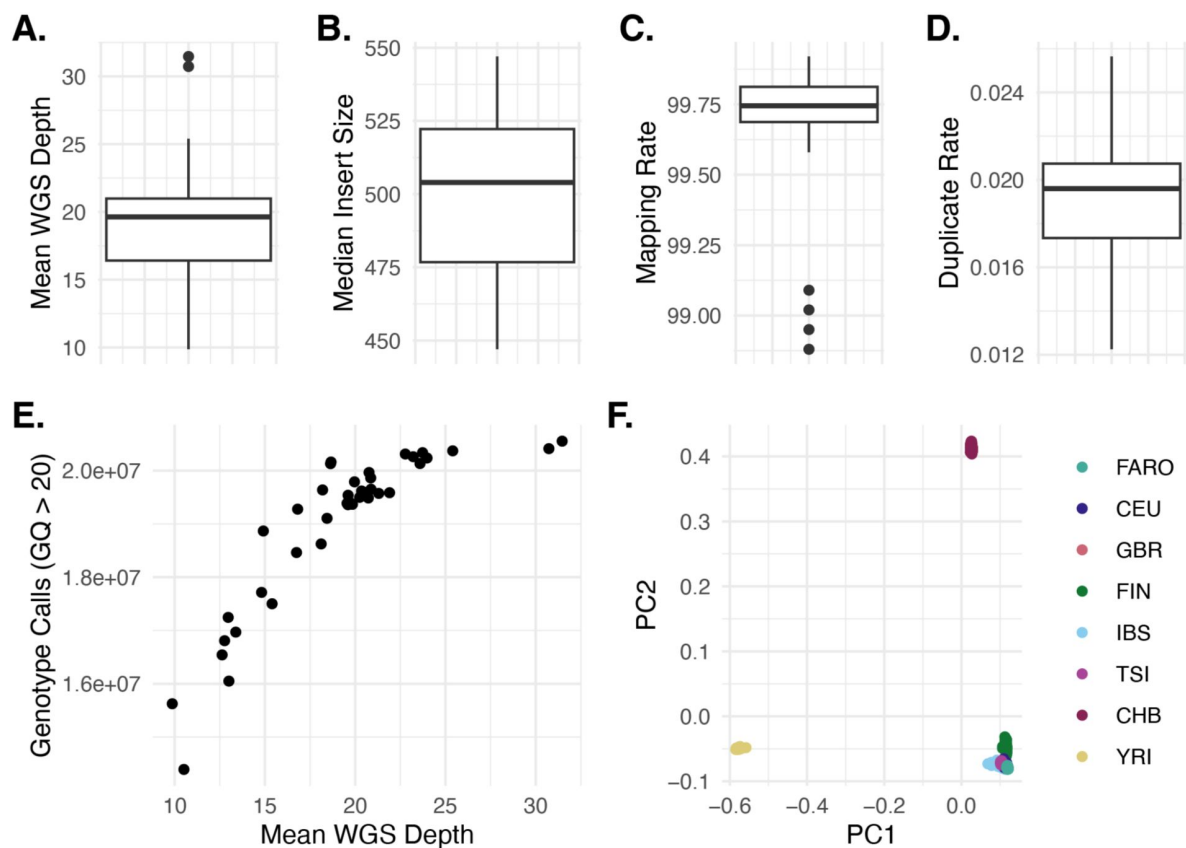
## Acknowledgements

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## Additional information

### Author Contributions

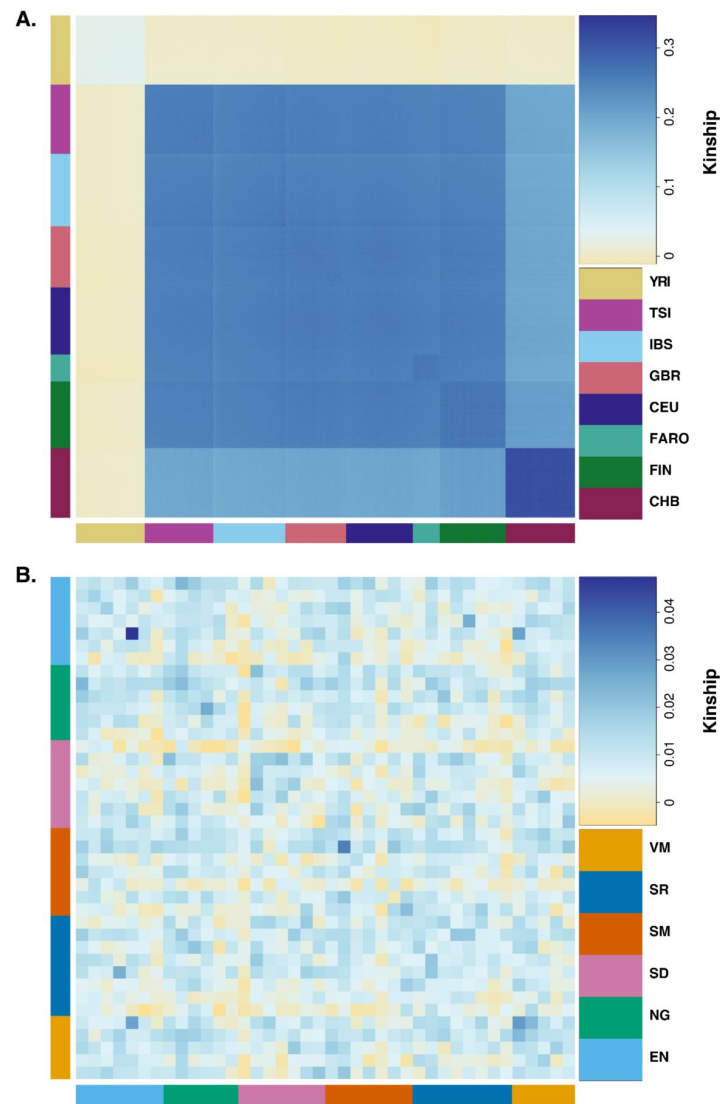
S.E.C., K.A.W., F.R. and N.O.G. designed the study. N.O.G. and K.D.A. secured ethical permissions, facilitated the inclusion of participants, and oversaw the compliance with ethical standards and protocols. G.A. prepared the data for the genealogy analysis. Ó.M. performed relatedness pruning and sample selection for WGS. L.N.L. prepared samples for WGS and performed the WGS analyses. A.E. processed sequencing data, performed imputation, and carried out quality control. I.H. conducted kinship, ROH, and selection scan analyses. A.R.M. carried out ancient admixture and local ancestry analysis. J.M. added functionalities to Haplonet software for semi-supervised



**Figure S1.**

### Quality control of whole genome sequencing data.

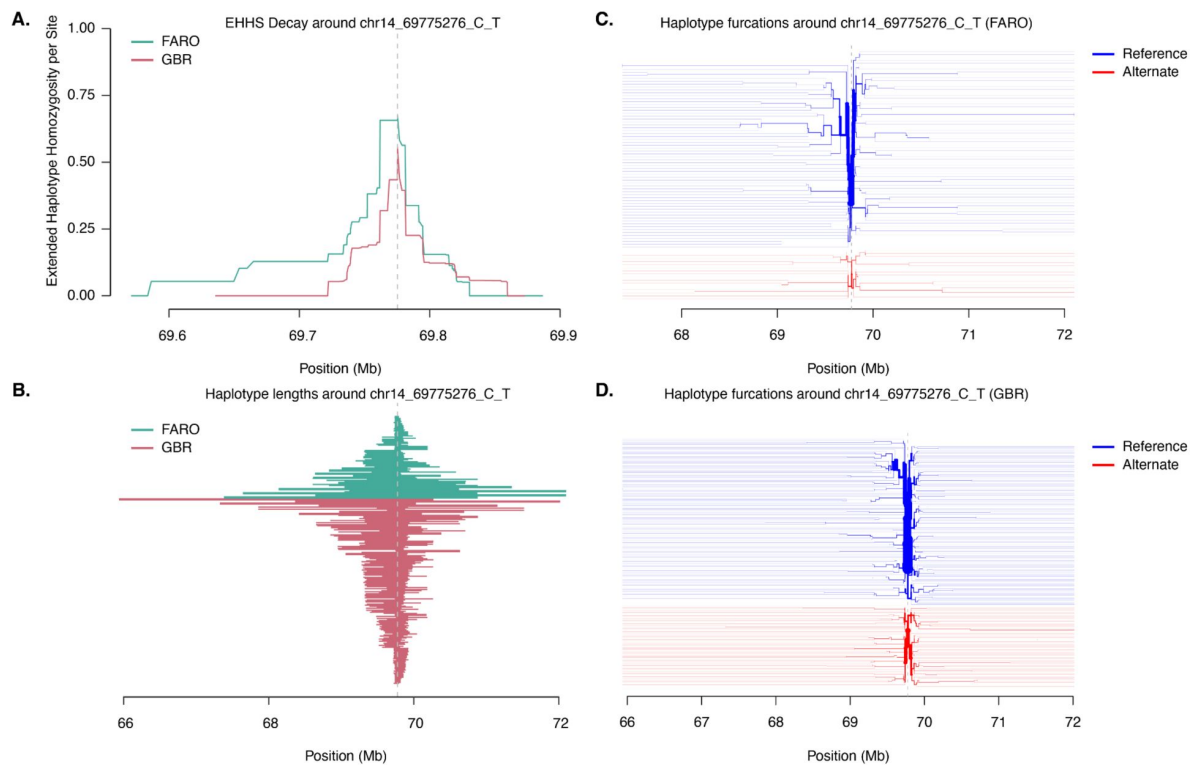
**A-D)** Distributions of QC metrics across Faroese samples sequenced (mean depth, median insert size, mapping rate, duplicate rate). **E)** Mean WGS depth versus number of genotype calls with GQ > 20. Singletons and calls with GT ≤ 20 were discarded before imputation. **F)** Principal component analysis (PCA) of Faroese genomes jointly called with relevant 10000 Genomes reference data captures African ancestry in the first component and East Asian ancestry in the second component (FARO, Faroese, CEU, Central Europeans, GBR, British, FIN, Finnish IBS, Iberian, TSI, Tuscan, CHB, Han Chinese, YRI, Yoruban).



**Figure S2.**

### Kinship matrices between modern populations.

**A)** Kinship estimated by *popkin*, including global reference populations from the 1000 Genomes and the Faroese WGS cohort (FARO). **B)** Kinship estimated by *popkin* for the Faroese WGS cohort only. The matrix is rescaled after subsetting the individuals, so although the scales are different, the overall structure remains the same. The matrices are symmetrical and ordered by population or region label as indicated by the colored bars along the rows and columns. The diagonal of each matrix is the estimated inbreeding coefficient. The Faroes region labels are: **VM** = Vágar and Mykines; **SR** = Suðuroy; **SM** = Suðurstreymoy; **SD** = Sandoy, Skúvoy, Stóra Dímun; **NG** = Norðoyggjar; and **EN** = Eysturoy og Norðstreymoy.

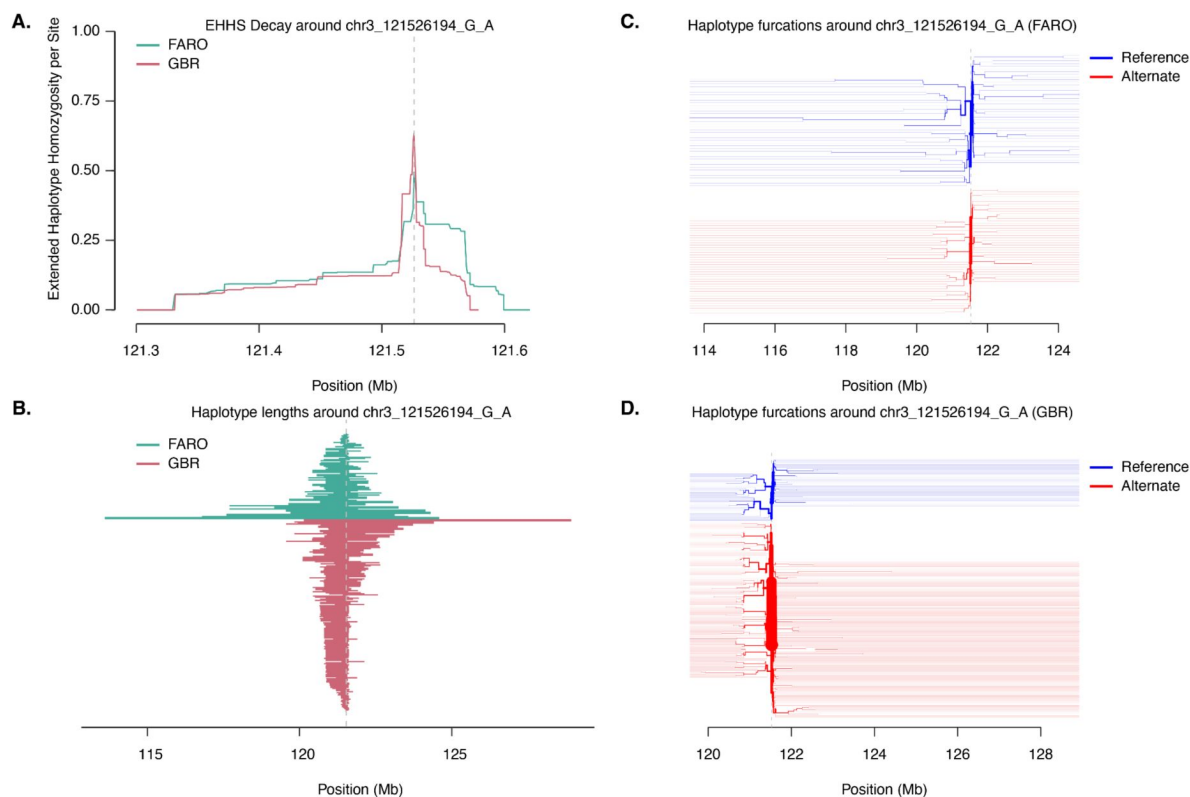


**Figure S3.**

### Haplotype visualizations for top XP-EHH variant in *SLC10A1* / *SRSF5* locus.

**A)** Decay in Expected Haplotype Homozygosity per Site (EHHS) for chr14\_69775276\_C\_T, comparing Faroese (FARO, teal) and British (GBR, red) haplotypes. **B)** Lengths for distinct haplotypes spanning chr14\_69775276\_C\_T comparing FARO (teal) and GBR (red). **C)** haplotype furcation plot for FARO centered on chr14\_69775276\_C\_T **D)** haplotype furcation for GBR centered on the same allele. In the haplotype furcation plots (panels **C** & **D**), haplotypes for the reference allele (C) are in blue, and those for the alternate allele (T) are in red.

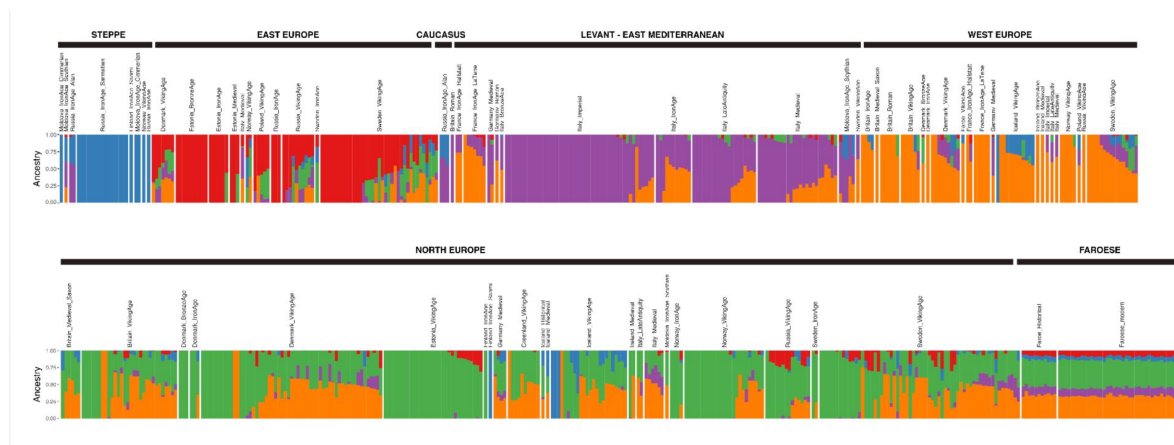




**Figure S4.**

#### Haplotype visualizations for top XP-EHH variant in *POLQ* locus.

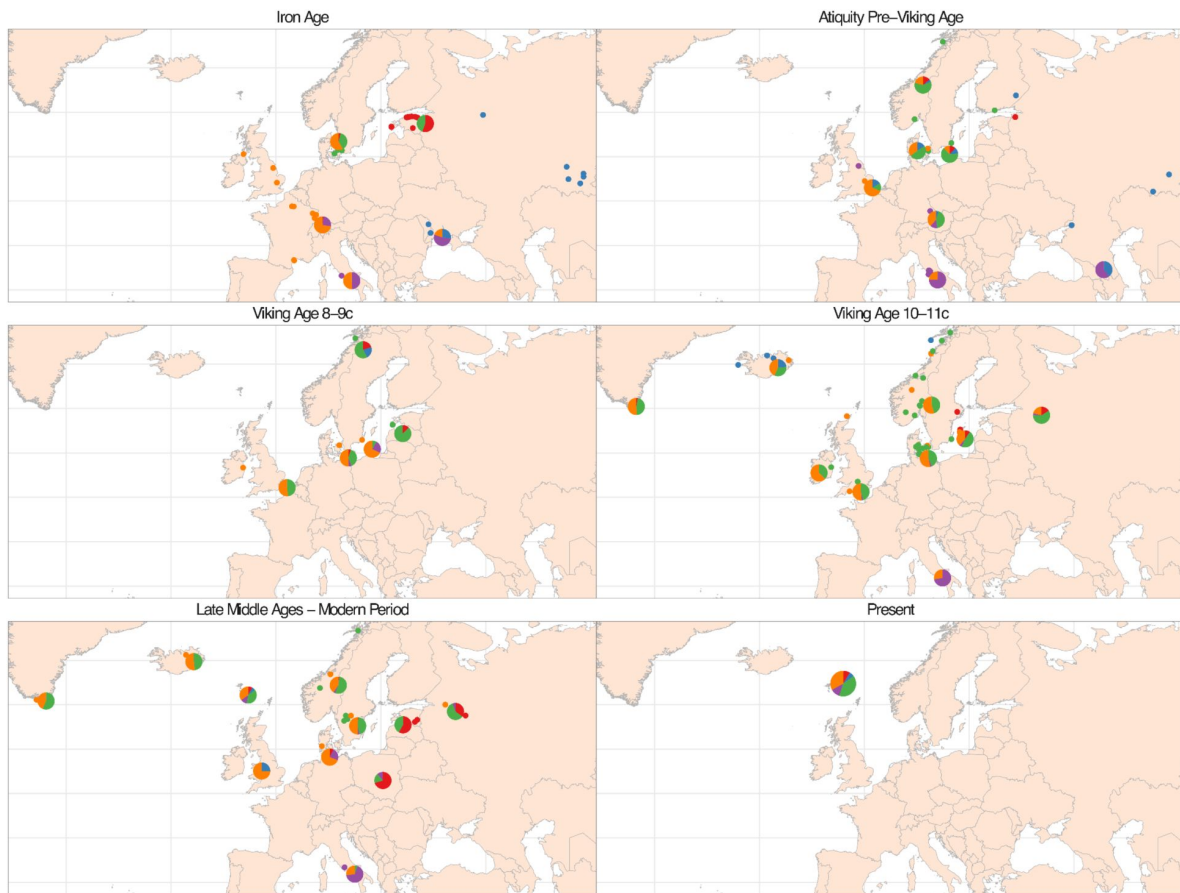
**A)** Decay in Expected Haplotype Homozygosity per Site (EHHS) for chr3\_121526194\_G\_A, comparing Faroese (FARO, teal) and British (GBR, red) haplotypes. **B)** Lengths for distinct haplotypes spanning chr3\_121526194\_G\_A comparing FARO (teal) and GBR (red). **C)** haplotype furcation plot for FARO centered on chr3\_121526194\_G\_A **D)** haplotype furcation for GBR centered on the same allele. In the haplotype furcation plots (panels **C** & **D**), haplotypes for the reference allele (G) are in blue, and those for the alternate allele (A) are in red.



**Figure S5.**

**Admixture plot showing proportions for 616 imputed ancient genomes from Europe together with 40 present-day Faroese genomes from this study.**

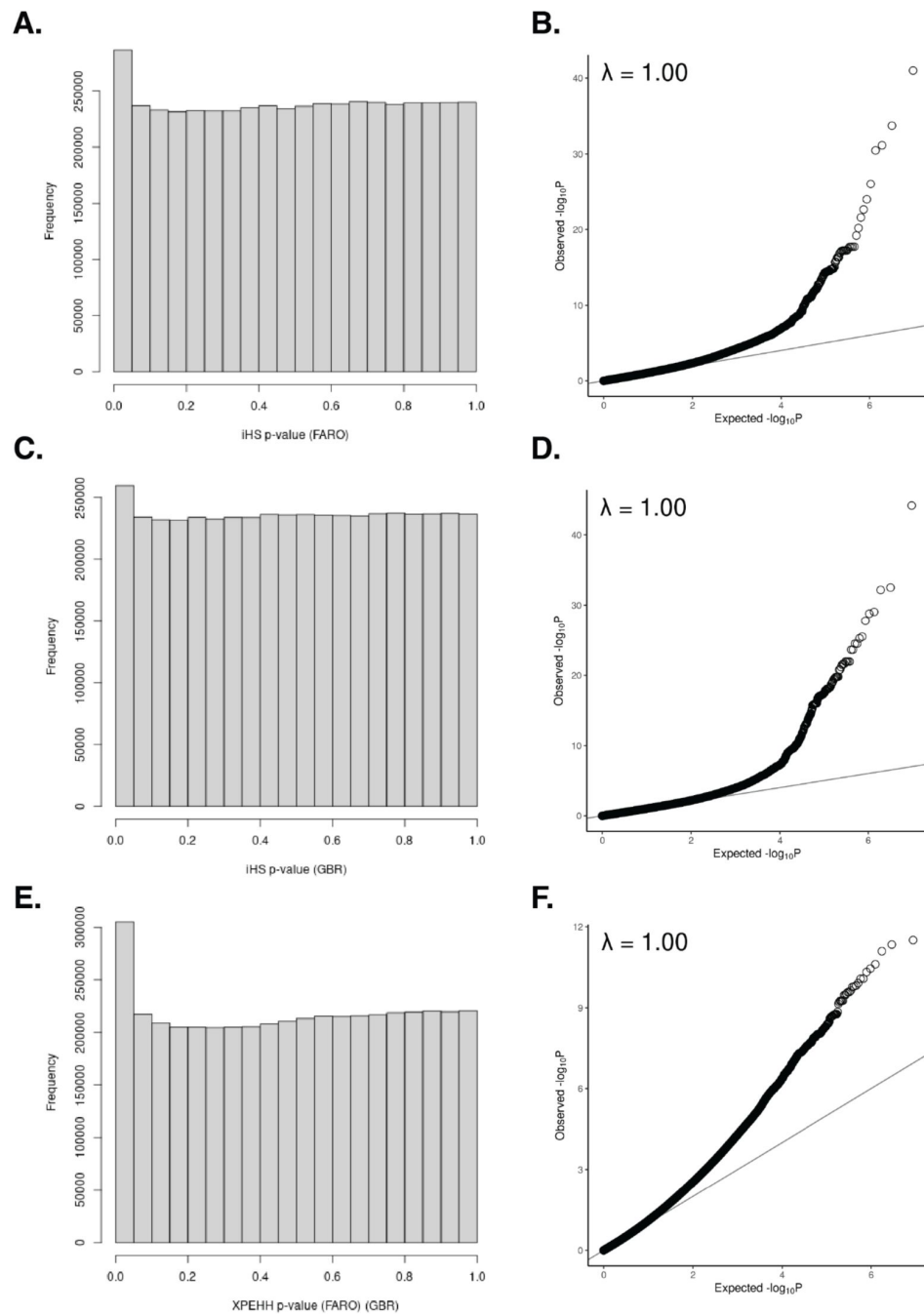
Ancient individual groups are categorized based on patterns of IBD clustering as inferred in Allentoft *et al.* 2024. The plot uses five colors to represent different ancestral sources, which are maximized in individuals in different regions of Europe: orange for “West Europe”, green for “North Europe”, blue for “Steppe”, purple for the “Levant and East Mediterranean” and red for “Eastern Europe”.



**Figure S6.**

**Map illustrating the geographical distribution of ancestry proportions for 616 ancient imputed individuals from Europe and 40 modern Faroese genomes from this study.**

The map is divided into panels to capture both geographical and temporal variations across Europe. Each pie chart on the map represents admixture proportions with five colors: green, orange, blue, red, and purple maximize Northern European, Celtic, Steppe, Eastern European, and Levant / East Mediterranean ancestries, respectively, reflecting different ancestral sources.



**Figure S7.**

### Q-Q plots and p-value histograms for selection statistics.

Histograms of p-value distributions for **A)** integrated haplotype score (iHS) in the Faroese (FARO) haplotypes, **C)** iHS in the British (GBR) haplotypes, **E)** cross-population expected haplotype homozygosity (XP-EHH) comparing FARO and GBR. Q-Q plots for observed versus expected log transformed p-values for **B)** iHS in FARO, **D)** iHS in GBR **F)** XP-EHH between FARO and GBR. The estimated lambda inflation value is shown in each Q-Q plot.

admixture and fine-structure analysis. S.E.C., N.M., and F.R. supervised analyses. I.H., S.E.C., and A.R.M. interpreted results and wrote the manuscript. All authors reviewed and contributed to the writing of this manuscript.

## Materials & Correspondence

Correspondence to N.O.G., S.E.C., and F.R..

## Additional files

**Supplementary Tables.** Table S1. Whole genome sequencing quality control metrics. Unique identifier (subject), sex, region, mean WGS depth (mean\_depth), mean depth on X chromosome (mean\_depth\_chrX), mean depth on Y chromosome (mean\_depth\_chrY), median insert size (insert\_median), mean absolute deviation of insert size (insert\_mad), contamination estimate (contamination), and number of genotype calls with GQ > 20 (non\_imputed\_calls). The Faroes region labels are: VM = Vágur and Mykines; SR = Suðuroy; SM = Suðurstreymoy; SD = Sandoy, Skúvoy, Stóra Dímun; NG = Norðoyggjar; and EN = Eysturoy og Norðstreymoy. Table S2. Faroese putatively functional alleles. Variants in the joint call set with CADD > 30 and at least two minor alleles observed in Faroese individuals, and no minor alleles observed in Finnish or Northern European reference individuals. Allele frequencies in Faroese individuals (af\_FARO), allele frequencies in 1000 Genomes British (af\_GBR), Central European (af\_CEU), and Finnish (af\_FIN), and allele frequencies in 1000G Europeans (af\_1000G\_EUR) and gnomAD (af\_GNOMAD) references. Table S3. HLA-B Allele frequencies. Counts of HLA-B alleles out of the total 80 haplotypes and mean quality scores in the Faroese WGS cohort, as well as HLA-B allele frequencies in three 1000 Genomes European cohorts (British [GBR], Central European [CEU], and Finnish [FIN]). HLA genotypes were determined using HLA\*LA. Table S4. Genomic regions with strong signals of positive selection. Genomic regions for the top 10 standardized iHS and XP-EHH values in the Faroese WGS cohort. We list the number of variants and all variant IDs in these regions which reach the p-value threshold corresponding to FDR < 0.01 (iHS:  $p < 2.72 \times 10^{-6}$ ; XPEHH:  $p < 2.35 \times 10^{-6}$ ). The regions are defined by the first and last variant position with a p-value below the cutoff for each particular statistic, as indicated in the “statistic” column. Protein-coding gene name(s), if any, with annotated variants reaching these thresholds are provided. We additionally provide the maximum absolute value of the given statistic (“max\_statistic”), its corresponding p-value and q-value (i.e. the minimum FDR when that particular test is considered significant), and the allele frequencies in the Faroese and British cohorts for the variant(s) with the maximum value. The variant(s) with the maximum value for each region are highlighted in bold font. Table S5. Admixture and ancient ancestry samples. Region, country, approximate age, publication source, and group label for ancient ancestry samples included in the ADMIXTURE and HaploNet analyses. [↗](#)

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**Reviewer #1 (Public review):**

Summary:

The paper reports an analysis of whole-genome sequence data from 40 Faroese. The authors investigate aspects of demographic history and natural selection in this population. The key findings are that the Faroese (as expected) have a small population size and are broadly of Northwest European ancestry. Accordingly, selection signatures are largely shared with other Northwest European populations, although the authors identify signals that may be specific to the Faroes. Finally, they identify a few predicted deleterious coding variants that may be enriched in the Faroes.

Strengths:

The data are appropriately quality-controlled and appear to be of high quality. Some aspects of the Faroese population history are characterized, in particular, by the relatively (compared to other European populations) high proportion of long runs of homozygosity, which may be relevant for disease mapping of recessive variants. The selection analysis is presented reasonably, although as the authors point out, many aspects, for example differences in iHS, can reflect differences in demographic history or population-specific drift and thus can't reliably be interpreted in terms of differences in the strength of selection.

#### Weaknesses:

The main limitations of the paper are as follows:

(1) The data are not available. I appreciate that (even de-identified) genotype data cannot be shared; however, that does substantially reduce the value of the paper. Minimally, I think the authors should share summary statistics for the selection scans, in line with the standard of the field.

(2) The insight into the population history of the Faroes is limited, relative to what is already known (i.e., they were settled around 1200 years ago, by people with a mixture of Scandinavian and British ancestry, have a small effective population size, and any admixture since then comes from substantially similar populations). It's obvious, for example, that the Faroese population has a smaller bottleneck than, say, GBR.

More sophisticated analyses (for example, ARG-based methods, or IBD or rare variant sharing) would be able to reveal more detailed and fine-scale information about the history of the populations that is not already known. PCA, ADMIXTURE, and HaplotNet analysis are broad summaries, but the interesting questions here would be more specific to the Faroes, for example, what are the proportions of Scandinavian vs Celtic ancestry? What is the date and extent of sex bias (as suggested by the uniparental data) in this admixture? I think that it is a bit of a missed opportunity not to address these questions.

(3) I don't really understand the rationale for looking at HLA-B allele frequencies. The authors write that "ankylosing spondylitis (AS) may be at a higher prevalence in the Faroe Islands (unpublished data), however, this has not been confirmed by follow-up epidemiological studies". So there's no evidence (certainly no published evidence) that AS is more prevalent, and hence nothing to explain with the HLA allele frequencies?

<https://doi.org/10.7554/eLife.107428.1.sa1>

#### Reviewer #2 (Public review):

In this paper, Hamid et al present 40 genomes from the Faroe Islands. They use these data (a pilot study for an anticipated larger-scale sequencing effort) to discuss the population genetic diversity and history of the sample, and the Faroes population. I think this is an overall solid paper; it is overall well-polished and well-written. It is somewhat descriptive (as might be expected for an explorative pilot study), but does make good use of the data.

The data processing and annotation follows a state-of-the-art protocol, and at least I could not find any evidence in the results that would pinpoint towards bioinformatic issues having substantially biased some of the results, and at least preliminary results lead to the identification of some candidate disease alleles, showing that small, isolated cohorts can be an efficient way to find populations with locally common, but globally rare disease alleles.

I also enjoyed the population structure analysis in the context of ancient samples, which gives some context to the genetic ancestry of Faroese, although it would have been nice if that



could have been quantified, and it is unfortunate that the sampling scheme effectively precludes within-Faroes analyses.

I am unfortunately quite critical of the selection analysis, both on a statistical level and, more importantly, I do not believe it measures what the authors think it does.

Major comments:

(1) Admixture timing/genomic scaling/localization:

As the authors lay out, the Faroes were likely colonized in the last 1,000-1,500 years, i.e., 40-60 generations ago. That means most genomic processes that have happened on the Faroese should have signatures that are on the order of  $\sim 1\text{-}2\text{cM}$ , whereas more local patterns likely indicate genetic history predating the colonization of the islands. Yet, the paper seems to be oblivious to this (to me) fascinating and somewhat unique premise. Maybe this thought is wrong, but I think the authors miss a chance here to explain why the reader should care beyond the fact that the small populations might have high-frequency risk alleles and the Faroes are intrinsically interesting, but more importantly, it also makes me think it leads to some misinterpretations in the selection analysis

(2) ROH:

Would the sampling scheme impact ROH? How would it deal with individuals with known parental coancestry? As an example of what I mean by my previous comment, 1MB is short enough in that I would expect most/many 1MB ROH-tracts to come from pedigree loops predating the colonization of the Faroes. (i.e, I am actually quite surprised that there isn't much more long ROH, which makes me wonder if that would be impacted by the sampling scheme).

(3) Selection scan:

We are talking about a bottlenecked population that is recently admixed (Faroese), compared to a population (GBR) putatively more closely related to one of its sources. My guess would be that selection in such a scenario would be possibly very hard to detect, and even then, selection signals might not differentiate selection in Faroese vs. GBR, but rather selection/allele frequency differences between different source populations. I think it would be good to spell out why XP-EHH/iHS measures selection at the correct time scale, and how/if these statistics are expected to behave differently in an admixed population.

(4) Similarly, for the discussion of LCT, I am not convinced that the haplotypes depicted here are on the right scale to reflect processes happening on the Faroes. Given the admixture/population history, it at the very least should be discussed in the context of whether the 13910 allele frequency on the Faroes is at odds with what would be expected based on the admixture sources.

(5) I am lacking information to evaluate the procedure for turning the outliers into p-values. Both iHS and XP-EHH are ratio statistics, meaning they might be heavy-tailed if one is not careful, and the central limit theorem may not apply. It would be much easier (and probably sufficient for the points being made here) to reframe this analysis in terms of empirical outliers.

(6) Oldest individual predating gene flow: It seems impossible to make any statements based on a single individual. Why is it implausible that this person (or their parents), e.g., moved to the Faroes within their lifetime and died there?

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**Author response:**

We thank the reviewers for their thoughtful comments and constructive suggestions. We describe how we will address each point below and are grateful for the guidance on areas where our work could be clarified or expanded. In particular, we note the following:

**Selection scan summary statistics:** In our revised manuscript, we will include summary statistics from the selection scans. We believe this addition will enhance transparency and provide additional context for readers.

**Reporting of outliers:** As highlighted by the editor, the reviewers expressed differing views on the most appropriate way to report outliers. To provide a comprehensive and balanced presentation, we will report both the empirical selection statistics and the corresponding converted p-values. This dual approach will allow readers to fully interpret the results under both perspectives.

**Methodological considerations:** We have carefully considered the reviewers' methodological suggestions and will incorporate them into our revisions where possible. These changes strengthen the rigor and clarity of the analyses.

**Public Reviews:****Reviewer #1 (Public review):***Summary:*

*The paper reports an analysis of whole-genome sequence data from 40 Faroese. The authors investigate aspects of demographic history and natural selection in this population. The key findings are that the Faroese (as expected) have a small population size and are broadly of Northwest European ancestry. Accordingly, selection signatures are largely shared with other Northwest European populations, although the authors identify signals that may be specific to the Faroes. Finally, they identify a few predicted deleterious coding variants that may be enriched in the Faroes.*

*Strengths:*

*The data are appropriately quality-controlled and appear to be of high quality. Some aspects of the Faroese population history are characterized, in particular, by the relatively (compared to other European populations) high proportion of long runs of homozygosity, which may be relevant for disease mapping of recessive variants. The selection analysis is presented reasonably, although as the authors point out, many aspects, for example differences in iHS, can reflect differences in demographic history or population-specific drift and thus can't reliably be interpreted in terms of differences in the strength of selection.*

*Weaknesses:*

*The main limitations of the paper are as follows:*

*(1) The data are not available. I appreciate that (even de-identified) genotype data cannot be shared; however, that does substantially reduce the value of the paper. Minimally, I think the authors should share summary statistics for the selection scans, in line with the standard of the field.*

We agree with the reviewer that sharing the selection scan results is important, so in the next revision of this manuscript we will make the selection scan summary statistics publicly

available, and clearly lay out the guidelines and research questions for which the data can be accessed.

*(2) The insight into the population history of the Faroes is limited, relative to what is already known (i.e., they were settled around 1200 years ago, by people with a mixture of Scandinavian and British ancestry, have a small effective population size, and any admixture since then comes from substantially similar populations). It's obvious, for example, that the Faroese population has a smaller bottleneck than, say, GBR.*

More sophisticated analyses (for example, ARG-based methods, or IBD or rare variant sharing) would be able to reveal more detailed and fine-scale information about the history of the populations that is not already known. PCA, ADMIXTURE, and HaplotNet analysis are broad summaries, but the interesting questions here would be more specific to the Faroes, for example, what are the proportions of Scandinavian vs Celtic ancestry? What is the date and extent of sex bias (as suggested by the uniparental data) in this admixture? I think that it is a bit of a missed opportunity not to address these questions.

We clarify that we did quantify the proportions of various ancestry components as estimated by HaploNet in main text Figure 5 and supplemental figures S5 and S6. In our revisions, we will include the average global ancestry of the various components in the Main Text so that this result is more clear.

We agree that more fine-scale demographic analyses would be informative. We have begun working on an estimation of the admixture date, for example, but have encountered problems with using different standard date estimation software, which give very inconsistent and unstable results. We suspect this might be due to the strong bottleneck experienced in the history of the Faroe Islands breaking one or more of the assumptions of these methods. We will continue working on this problem in coming months, possibly using simulations to assess where the problem might be. We recognize that our relatively small sample size places limits on the fine-scale demographic analyses that can be performed. We are addressing this in ongoing work by generating a larger cohort, which we hope will enable more detailed inference in the future.

*(3) I don't really understand the rationale for looking at HLA-B allele frequencies. The authors write that "ankylosing spondylitis (AS) may be at a higher prevalence in the Faroe Islands (unpublished data), however, this has not been confirmed by follow-up epidemiological studies". So there's no evidence (certainly no published evidence) that AS is more prevalent, and hence nothing to explain with the HLA allele frequencies?*

We agree that no published studies have confirmed a higher prevalence of ankylosing spondylitis (AS) in the Faroe Islands. Our recruitment data suggest that AS might be more common than in other European populations, but we understand that this is only based on limited, unpublished observations and what we are hearing from the community. We emphasized in our original manuscript that this is based on observational evidence from the FarGen project. However, as this reviewer pointed out, we can be more clear that this prevalence has not been formally studied.

In our next revision we will clarify in the text that our recruitment data suggest a higher prevalence of AS may be possible, but more formal epidemiological studies are needed to confirm this observation. The reason we study HLA-B allele frequencies is to see if the genetic background of the Faroese population could help explain this possible difference, since HLA-B27 is already known to play a strong role in AS.

## Reviewer #2 (Public review):

*In this paper, Hamid et al present 40 genomes from the Faroe Islands. They use these data (a pilot study for an anticipated larger-scale sequencing effort) to discuss the population genetic diversity and history of the sample, and the Faroes population. I think this is an overall solid paper; it is overall well-polished and well-written. It is somewhat descriptive (as might be expected for an explorative pilot study), but does make good use of the data.*

*The data processing and annotation follows a state-of-the-art protocol, and at least I could not find any evidence in the results that would pinpoint towards bioinformatic issues having substantially biased some of the results, and at least preliminary results lead to the identification of some candidate disease alleles, showing that small, isolated cohorts can be an efficient way to find populations with locally common, but globally rare disease alleles.*

*I also enjoyed the population structure analysis in the context of ancient samples, which gives some context to the genetic ancestry of Faroese, although it would have been nice if that could have been quantified, and it is unfortunate that the sampling scheme effectively precludes within-Faroes analyses.*

We note that although the ancestry proportions are not specified in the main text, we did quantify ancestry proportions in the modern Faroese individuals and other ancient samples, and we visualized these proportions in Figure 5 and Supplementary Figures S5 and S6. As stated in our response to Reviewer #1, in our revisions, we will more clearly state the average global ancestry of the various components in the Main Text.

*I am unfortunately quite critical of the selection analysis, both on a statistical level and, more importantly, I do not believe it measures what the authors think it does.*

*Major comments:*

*(1) Admixture timing/genomic scaling/localization:*

*As the authors lay out, the Faroes were likely colonized in the last 1,000-1,500 years, i.e., 40-60 generations ago. That means most genomic processes that have happened on the Faroese should have signatures that are on the order of ~1-2cM, whereas more local patterns likely indicate genetic history predating the colonization of the islands. Yet, the paper seems to be oblivious to this (to me) fascinating and somewhat unique premise. Maybe this thought is wrong, but I think the authors miss a chance here to explain why the reader should care beyond the fact that the small populations might have high-frequency risk alleles and the Faroes are intrinsically interesting, but more importantly, it also makes me think it leads to some misinterpretations in the selection analysis*

See response to point #3

*(2) ROH:*

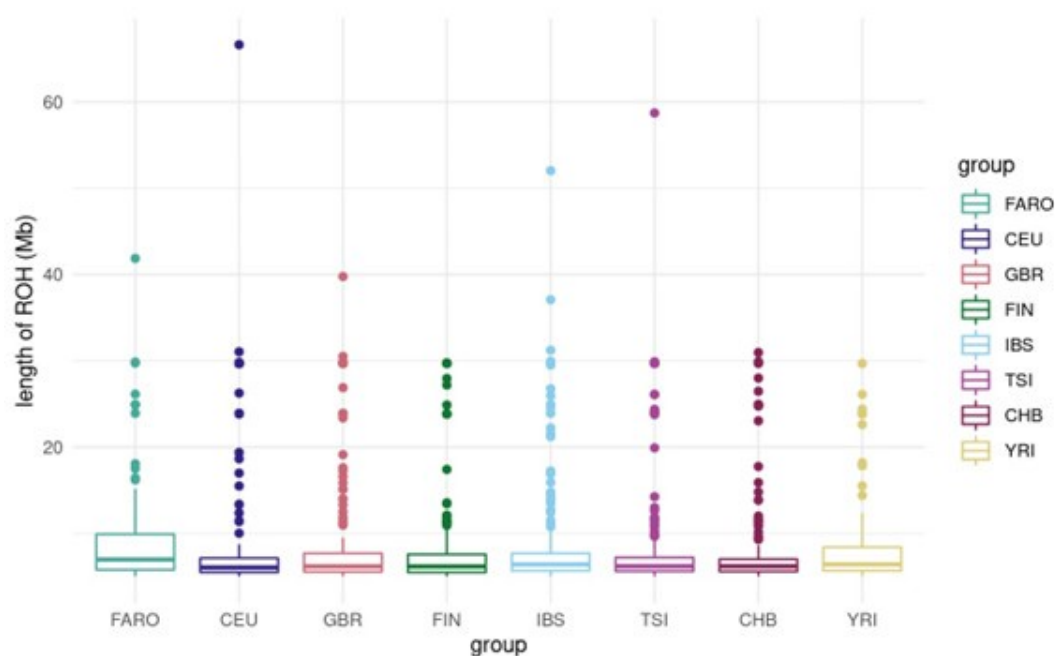
*Would the sampling scheme impact ROH? How would it deal with individuals with known parental coancestry? As an example of what I mean by my previous comment, 1MB is short enough in that I would expect most/many 1MB ROH-tracts to come from pedigree loops predating the colonization of the Faroes. (i.e, I am actually quite surprised that there isn't much more long ROH, which makes me wonder if that would be impacted by the sampling scheme).*

The sampling scheme was designed to choose 40 Faroese individuals that were representative of the different regions and were minimally related. There were no pairs of third-degree relatives or closer ( $\pi\text{-hat} > 0.125$ ) in either the Faroese cohort or the reference populations. It is possible that this sampling scheme would reduce the amount of longer ROHs in the population, but we should still be able to see overall patterns of ROH reflective of bottlenecks in the past tens of generations. Additionally, based on this reviewer's earlier comment, 1 Mb ROHs would still be relevant to demographic events in the last 40-60 generations given that on average 1 cM corresponds to 1 Mb in humans, though we recognize that is not an exact conversion.

That said, the "sum total amount of the genome contained in long ROH" as we described in the manuscript includes all ROHs greater than 1Mb. Although we group all ROHs longer than 1Mb into one category in the current manuscript, we can look more specifically at the distribution of the longer ROH in future revisions and add discussion into what this might tell us about the timing of bottlenecks.

For now, we share a plot of the distribution in ROH lengths across all individuals for each cohort. As this plot shows, there certainly are ROHs longer than 1Mb in the Faroese cohort, and on average there is a higher proportion of long ROH particularly in the 5-15 Mb range in the Faroese cohort relative to the other cohorts.

#### Author response image 1.



#### (3) Selection scan:

*We are talking about a bottlenecked population that is recently admixed (Faroese), compared to a population (GBR) putatively more closely related to one of its sources. My guess would be that selection in such a scenario would be possibly very hard to detect, and even then, selection signals might not differentiate selection in Faroese vs. GBR, but rather selection/allele frequency differences between different source populations. I think it would be good to spell out why XP-EHH/iHS measures selection at the correct time scale, and how/if these statistics are expected to behave differently in an admixed population.*

The reviewer brings up good points about the utility of classical selection statistics in populations that are admixed or bottlenecked, and whether the timescale at which these statistics detect selection is relevant for understanding the selective history of the Faroese population. We break down these concerns separately.

(1) Bottlenecks: Recent bottlenecks result in higher LD within a population. However, demographic events such as bottlenecks affect global genomic patterns while positive selection is expected to affect local genomic patterns. For this reason, iHS and XP-EHH statistics are standardized against the genome-wide background, to account for population-specific demographic history.

(2) Admixture: The term “admixture” has different interpretations depending on the line of inquiry and the populations being studied. Across various time and geographic scales, all human populations are admixed to some degree, as gene flow between groups is a common fixture throughout our history. For example,

even the modern British population has “admixed” ancestry from North / West European sources as well, dating to at least as recently as the Medieval & Viking periods (Gretzinger et al. 2022, Leslie et al. 2015), yet we do not commonly consider it an “admixed” population, and we are not typically concerned about applying haplotype-based statistics in this population. This is due to the low divergence between the source populations. In the case of the Faroe Islands, we believe admixture likely occurred on a similar timescale. We see low variance in ancestry proportions estimated by HaploNet, both from the historical Faroese individuals (250BP) and the modern samples. This indicates admixture predating the settlement of the Faroe Islands, where recombination has had time to break up long ancestry tracts and the global ancestry proportions have reached an equilibrium. That is, these ancestry patterns suggest that the modern Faroese are most likely descended from already admixed founders. We mention this as a likely possibility in the main text: “This could have occurred either via a mixture of the original “West Europe” ancestry with individuals of predominantly “North Europe” ancestry, or a by replacement with individuals that were already of mixed ancestry at the time of arrival in the islands (the latter are not uncommon in Viking Age mainland Europe).” And, as with the case of the British population, the closely-related ancestral sources for the Faroese founders were likely not so diverged as to have differences in allele frequencies and long-range haplotypes that would disrupt signals of selection from iHS or XP-EHH.

(3) Time scale: It is certainly possible, and in fact likely, that iHS measures selection older than the settlement of the Faroe Islands. In our manuscript, we calculated iHS in both the Faroese and the closely related British cohort, and we highlight in the main Main Text that the top signals, with the exception of LCT, are shared between the two cohorts, indicative of selection that began prior to the population split. iHS is a commonly calculated statistic, and it is often calculated in a single population without comparing to others, so we feel it is important to show our result demonstrating these shared selection signals. In future revisions, we will emphasize in the main text that we are not claiming to have identified selection that occurred in the Faroese population post-settlement with the iHS statistic. As far as XP-EHH, it is a statistic designed to identify differentiated variants that are fixed or approaching fixation in one population but not others. The time-scale of selection that XP-EHH can detect would therefore be dependent on the populations used for comparison. As XP-EHH has the best power to identify alleles that are fixed or approaching fixation in one population but not others, it is less likely to detect older selection events / incomplete sweeps from the source populations.

In our next revision, we will more clearly state limitations of these statistics under various population histories, and clarify the time-scale at which we are detecting selection for iHS vs XP-EHH.



*(4) Similarly, for the discussion of LCT, I am not convinced that the haplotypes depicted here are on the right scale to reflect processes happening on the Faroes. Given the admixture/population history, it at the very least should be discussed in the context of whether the 13910 allele frequency on the Faroes is at odds with what would be expected based on the admixture sources.*

We agree that more investigation into the LCT allele frequency in the other ancient samples may provide some insight into the selection history, particularly in light of ancient admixture. Please note, we did look at the allele frequency of the LCT allele rs4988235 and stated in the main text that it was present at high frequencies in the historical (250BP) Faroese samples. The frequency of this allele in the imputed historical Faroese samples is 82% while the allele is present at ~74% frequency in modern samples. We did not report the exact percentage in the main text because the sample size of the historical samples (11 individuals) is small and coverage of ancient samples is low, leading to potential errors in imputation. However, we can try to calculate the LCT allele frequency in other ancient samples, and assuming that we have good proxies for the sources at the time of admixture, we may calculate the expected allele frequency in the admixed ancestors of the Faroese founders in the next revision.

*(5) I am lacking information to evaluate the procedure for turning the outliers into p-values. Both iHS and XP-EHH are ratio statistics, meaning they might be heavy-tailed if one is not careful, and the central limit theorem may not apply. It would be much easier (and probably sufficient for the points being made here) to reframe this analysis in terms of empirical outliers.*

Given that there are disagreements on the best approach to reporting selection scan results from the reviewers, in our revision, we can additionally supply both the standardized iHS / XP-EHH values in the supplementary information as well as these values transformed to p-values. As the p-values are derived from the empirical distribution, the “significant” p-values are also empirical outliers from the empirical distribution, so the conclusions of the manuscript do not change. We found that the p-value approach and controlling for FDR is more conservative, with fewer signals reaching “significance” than are considered empirical outliers based on common approaches such as IQR or arbitrary percentile cutoffs.

*(6) Oldest individual predating gene flow: It seems impossible to make any statements based on a single individual. Why is it implausible that this person (or their parents), e.g., moved to the Faroes within their lifetime and died there?*

We agree with the reviewer that this is a plausible explanation, and in future revisions we will update the main text to acknowledge this possibility.

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